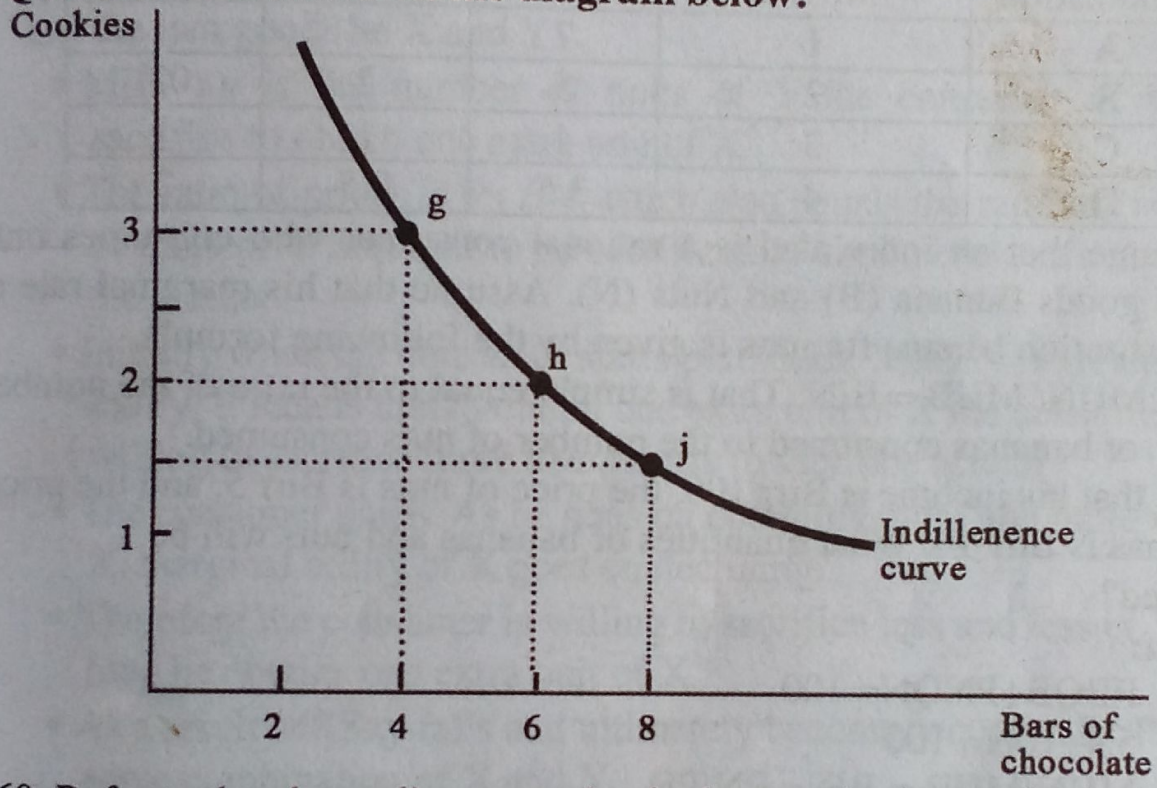
59. Indifference curve is convex to the origin. When the indifference curve be a straight line?

***Explanation*:**

- If two goods are perfect substitutes for a consumer, the ratio of the marginal utilities that the two goods provide is constant.
- Indifference curves, with slope MU1/MU2 are straight lines, since the marginal rate of substitution is constant

Question 60 is based on the diagram below:



60. Refer to the above diagram, what is the marginal rate of substitution between g and h?

***Solution*:**

MRS= change in cookies/ change in Bars of chocolate

MRS= 2-3/6-4

MRS= ½

61. If a consumer's marginal rate of substitution equals 2 eggs for1 bread,

A. the consumer's indifference curve must be positively sloped.

B. the consumer's indifference curve must be convex with respect to the origin of the graph.

C. the ratio of the consumer's marginal utility of 1 egg to that of 1 bread must equal ½.

D. all of the above are true.

***Solution*:**

MRS EB = change in bread/ change in egg

MRSEB = ½ **Answer: D**

62. Given the data below, compute the MRS of movies for concert and vice versa.

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Combination	A	B	C	D
Movies per month	1	2	3	4
Concerts per month	7	5	4	3½

***Solution*:**

MRS mc= change in concert/ change in movies

MRS cm = change in movies / change in concert

Combination	Movies	Concert	MRSmc	MRScm
A	1	7	-	-
B	2	5	2	0.5
C	3	4	1	1
D	4	3.5	0.5	2

63. Assume that an individual is a rational consumer who consumes only two goods Banana (B) and Nuts (N). Assume that his marginal rate of substitution banana for nuts is given by the following formula.

MRS = MUN/ MUB = B/N. That is simply equal to the ratio of the number of bananas consumed to the number of nuts consumed.

Assume that his income is Birr 100, the price of nuts is Birr 5, and the price of bananas is Birr 10. What quantities of bananas and nuts will be consumed?

***Solution*:**

PBQB+PNQN= 100

5N+10N= 100

MUN/MUB = B/N= PN/PB

5/10 = B/N

N= 10B/5 = N =2B

5N+10B = 100

5(2B)+ 10B= 100

10B+10B = 100

20B = 100

B= 5

N = 2B = 2x5 = 10

The person consumes 5 units of banana and 10 units of nuts.

64. A consumer consumes only two goods X and Y. Her money income is Birr 24 and the prices of Goods X and Y are Birr 4 and Birr 2 respectively.

a. Can the consumer afford a bundle 4X and 5Y?

b. What will be the MRSxy when the consumer is in equilibrium?

***Explanation*:**

Given: income = 24, Px =4 , Py = 2

a. QxPx + QyPy = Income

The cost of 4X + 5Y = (4x4) + (5x2) = Birr 26

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Since the income is only Birr 24 the consumer cannot afford the bundle.

b. When the consumer is in equilibrium,

MRS = Px/Py

Substituting Px = 4, and Py = 2,

MRS = 4/2 = 2

65. For a consumer to be in equilibrium why must marginal rate of substitution be equal to the ratio of prices of the two goods?

Explanation:

Let the two goods be X and Y.

- MRS xy is the number of units of Y the consumer is willing to sacrifice to obtain one extra unit of X.
- The ratio of prices is Px /Py which also equals the ratio of the number of units of Y required to be sacrificed to obtain one extra unit of X in the market.
- Initially when the consumer starts purchases, MRSxy is greater than Px/Py. It means that to obtain one extra unit of X the consumer is willing to sacrifice more than he has to sacrifice actually.
- The consumer gains. As he goes on obtaining more and more units of X, marginal utility of X goes on declining.
- Therefore the consumer is willing to sacrifice less and less of Y each time he obtains one extra unit of X.
- As a result MRSxy falls and ultimately becomes equal to Px/Py at some combination of X and Y.
  At this combination the consumer is in equilibrium.
- If the consumer attempts to obtain more units of X beyond the equilibrium level, MRSxy will become less than Px/Py and he will start losing. So he will not try to obtain more of X.

3.5 The budget line:

The budget line represents the maximum combination of goods and services that the consumer can purchase from the market given his limited income and the prices for goods and services.

By assuming that a consumer spends all of his or her incomes on good *X* and on good *Y* we can express the budget constraint as

$$PXQX + PYQY = I$$

Where *PX* is the price of good *X*, *QX* is the quantity of good *X*, *PY* is the price of good *Y*, *QY* is the quantity of good *Y*, and *I* is the consumer's money income

The maximum affordable combination of products that a consumer can afford is shown by the budget line

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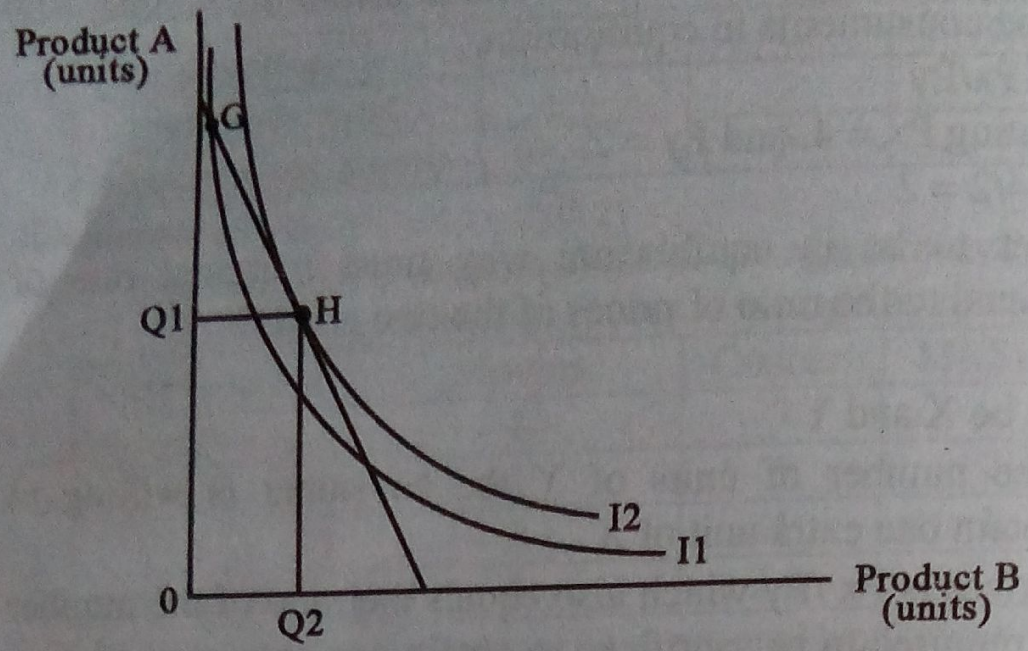

Figure 3.3: the budget line.

A consumer will maximize utility by consuming on the highest possible indifference curve A consumer could consume at G, for example, but would be on a higher indifference curve at H.

This means that to maximize utility the consumer would consume Q1 of product A and Q2 of product B.

The slope of the budget line depends on the relative prices of the products. It is given by:

$$\frac{Price\ of\ B}{Price\ of\ A} = \frac{PB}{PA}$$

The consumer is maximizing utility where the budget line and indifference curve are tangent i.e.

$$\frac{MUB}{MUA} = \frac{PB}{PA} \text{ (at point H)}$$

Illustrative Example

66. Holding all else equal, indicate how each of the following changes would affect a budget constraint that limits consumption of goods (Y) and services (X).

A. Deflation that uniformly drops the price of all goods and services.

***Explanation*:**

- Deflation that drops the price of all goods and services results in a parallel rightward (outward) shift in the budget constraint.
- Holding income constant, lower prices make it possible for consumers to buy more goods and services with the same total amount of spending

B. Inflation that consistently increases the price of all goods and services.

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Explanation:

- Inflation that increases the price of all goods and services results in a parallel leftward (inward) shift in the budget constraint.
- Holding income constant, higher prices reduce the amount of goods and services that consumers can buy with a fixed amount of spending

67. The absolute value of the slope of the budget constraint is equal to
    A. the price of good on the vertical axis divided by the price of the good on the horizontal axis.
    B. the marginal rate of substitution between the two goods in question.
    C. the price of good on the horizontal axis divided by the price of the good on the vertical axis.
    D. quantity of the good on the vertical axis divided by the quantity of the good on the horizontal axis.

Explanation:

The slope of the budget line is equal to the ratio of the prices of the two goods

Slope of the budget line = PX/PY **Answer: C**

68. The alternative combinations of two goods which a consumer can purchase with a given money income is:
    A. a production possibilities curve.
    B. a demand curve.
    C. consumer equilibrium.
    D. a budget line

Explanation: Budget line shows the maximum combination of two goods that is possible for a consumer to buy, given a budget constraint and the market prices of the two goods. **Answer: D**

69. A budget line relating to good A and good B has intercepted of 60 units of good A and 40 units of good B. if the price of good A is Birr 12. Find out
    A. The consumer income
    B. What is the price of good B?
    C. what is the slope of the budget line.

Solution:

A. Intercept of X- axis = Income/PA

60 = I/ 12

Income = 720

B. 40 = 720/PB

PB = 720/40

PB = 18

C. Slope of the budget line = PA/PB

= 12/18

= 0.67

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70. A consumer spends all her income on food and cloth. If the price of food is Birr 5 and price of cloth is 15. A consumer maximizes by purchasing of 30 units of food and 60 units of cloth.

Find out:

A. consumer income B. slope of the budget line

Solution:

A. PF X QF + PC x QC = Income

I = 5x30 + 15x60

I= 1050

B. Slope of the budget line = PF/PC

= 5/15

= 1/3

71. When a consumer's income increases:

A. The slope of the budget line changes

B. The budget line of the consumer shifts to the left

C. Purchasing power of the consumer's income is not affected

D. If the two goods consumed are normal, demand for both goods increases

Explanation:

- Income effect is said to be negative when with the increase in income the consumer reduces consumption of the good. Such goods for which income effect is negative are inferior goods.
- In the case of normal goods with an increase in income the consumer increase the consumption of the goods.

Answer: D

72. A consumer has Birr 100 per day to spend on product A, which has a unit price of Birr 7, and product B, which has a unit price of Birr 15. What is the slope of the budget line if good A is on the horizontal axis and good B is on the vertical axis?

Solution: Slope of the budget line = price of A/ price of B

= PA/PB

= 7/15

73. When income of the consumer rises, the budget line moves:

A. Inwards by the same proportion as the increase in purchasing power

B. Outwards by the same proportion as the increase in purchasing power

C. Outwards by proportion less than the increase in purchasing power

D. Outwards by proportion more than the increase in purchasing power.

Explanation:

- When income of the consumer rises, the budget line moves outwards by the proportion of increase in the purchasing power.
- While when income declines the budget line moves inwards in accordance with the decrease in the purchasing power.

Answer: B

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74. What is a budget line? What does the point on it indicate in terms of prices?

Explanation:

- Budget line is the locus of points that show different possible combinations of the two goods which a consumer can afford, given his income and the market prices of the two goods.
- In terms of prices, a point on the budget line represents the ratio of price of the good shown on the X axis to the price of the good shown on the Y-axis.

3.6. Effect of change in income:

An increase in income will shift the budget line upward (and rightward). A decrease in income will shift the budget line downward (and leftward). These shifts are parallel changes in income do not affect the budget line's slope

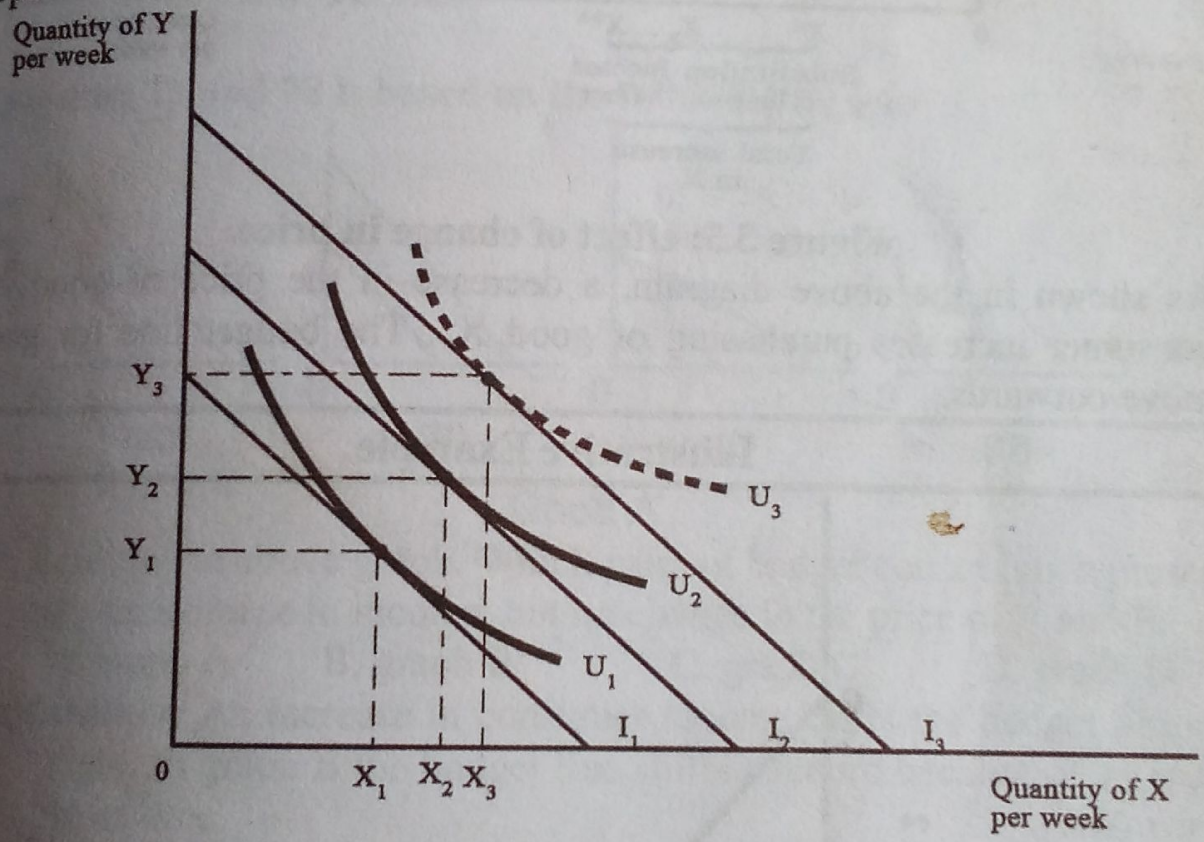

Figure 3. 4: Effects of Increasing Income on Quantities of X and Y Chosen

- In response to the increase in income the quantity of X purchased increases from X_1 to X_2 and X_3 while the quantity purchased of Y also increases from Y_1 to Y_2 to Y_3.

Effect of change in price:

- A change in the price of one good causes both the slope and an intercept of the budget line to change.
- The change also involves moving to a new utility-maximizing choice on another indifference curve with a different MRS.

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The quantity demanded of the good whose price has changed changes

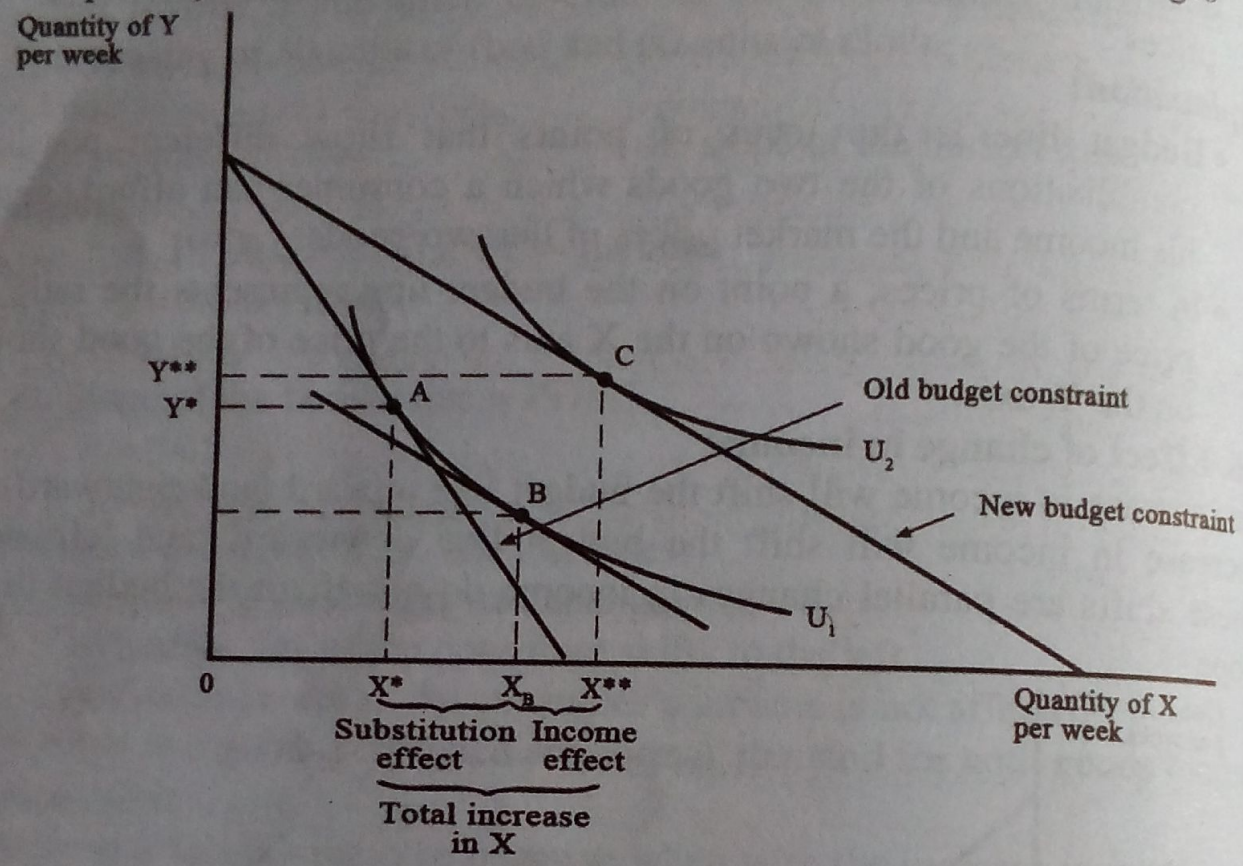
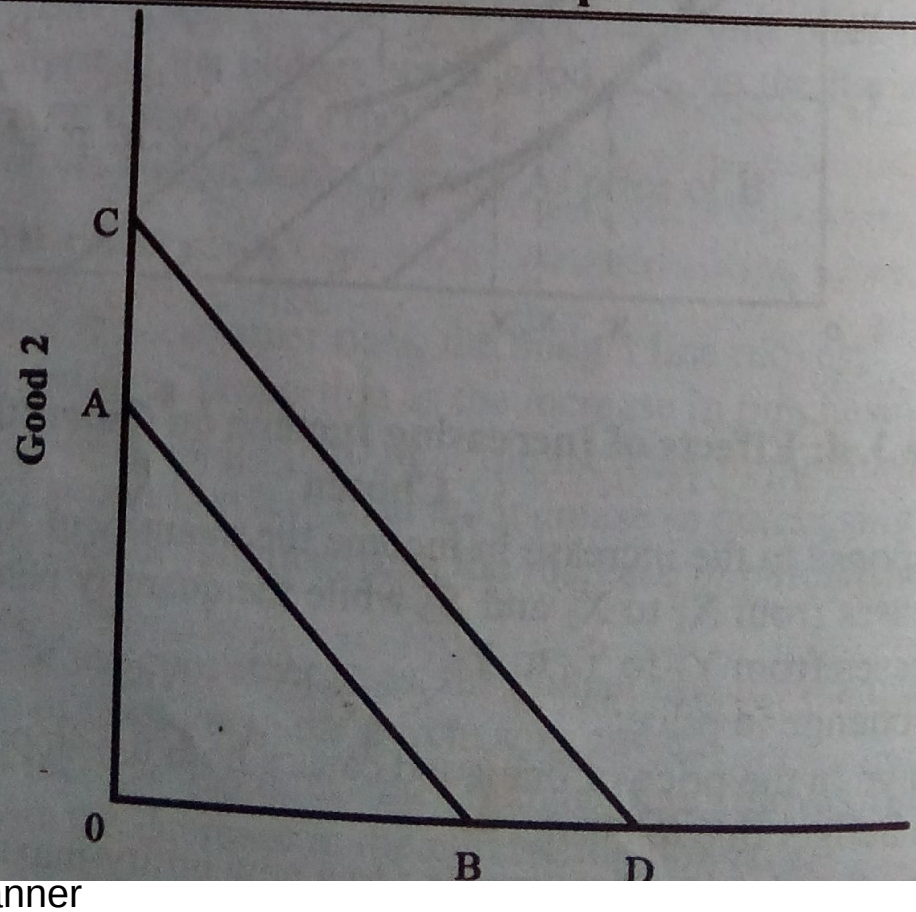


Figure 3.5: effect of change in price.

As shown in the above diagram, a decrease in the price of good X the consumer increases purchasing of good X . The budget line for good X move outwards.

Illustrative Example



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75. Refer to the above graph. The shift of the budget line from AB to CD is consistent with:
A. a decrease in money income. C. an increase in the price of Good 1.
B. an increase in money income. D. a decrease in the price of Good 2.

Explanation:
Income changes will cause parallel shift of the budget line outward. The budget line from AB shifts to CD when there is an increase in consumers income.

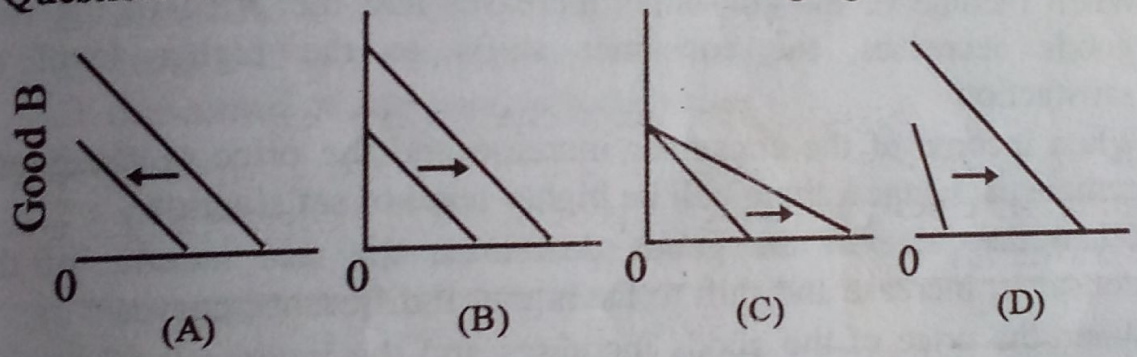
Answer: B

76. Refer to the above graph. The shift of the budget line from CD to AB is consistent with:
A. a decrease in money income. C. an increase in the price of Good 1.
B. an increase in money income D. a decrease in the price of Good 2.

Explanation:
A shift of the budget line from CD to AB is consistent with a decrease in consumers' income.

Answer: A

Question 77 and 78 is based on the following figure:



77. Refer to the above graph. Which pairs of budget constraints represent only an increase in income, but no change in the price of A and B?
A. graph A B. graph B C. graph C D. graph D

Explanation: An increase in consumer income shifts the budget line to the right. In graph B the budget line shifts outward because of an increase in income.

Answer: B

78. Refer to the above graph. Which pairs of budget constraints represent only a decrease in the price of Good A, but no change in income?
A. graph A B. graph B C. graph C D. graph D

Explanation: When there is a decrease in the price of good A, the purchasing power of the consumer increase. Consumers can buy more of good A and the budget line move outward.

Answer: C

79. Assuming the two goods consumed are normal goods, a decrease in the income of the consumer will move:
A. to a lower indifference curve
B. The consumer to higher indifference curve

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C. Upward along the original (old) indifference curve
D. Downward along the original (old) indifference curve

***Explanation*:**
For normal goods, an increase in income of the consumers will shift to the higher indifference curve. A decrease in income will shift to lower indifference curve.

Answer: A

80. When do the following changes not lead to a higher level of satisfaction:
A. The income of the consumer is doubled and the price levels are halved
B. The price of the two goods consumed decrease
C. The income of the consumer is increased and the prices of the goods remain unchanged
D. The price of all goods increase and the income level remain the same.

***Explanation*:**
- When income of the consumer increases and the price of the two goods decreases, the consumer shifts to the higher level of satisfaction
- When income of the consumer increase and the price of the goods remain unchanged, there will be higher level of satisfaction
- When the price of the goods decreases, the real income of the consumer increase and shift to the higher indifference curves
- When the price of the goods increases and the income level remain the same, the purchasing power of the consumer decreases. This lead to a low level of satisfaction.

Answer: D

81. What is the impact of a price falls for consumption inferior and normal goods?

***Explanation*:**
- When the price of a good falls, we are better off and we are likely to buy more of those good and other goods if they are normal goods.
- When income raises demand for normal good increases.
- When income raises demand for inferior goods decreases.

82. The income effect of a reduction in the price of a normal good tends to:
A. Reduce the quantity demanded by the consumer
B. Increase the quantity demanded by the consumer
C. Encourage the consumer to substitute the good for other
D. Keep the quantity demanded of the good unchanged

***Explanation*:** Income effect is the change in the quantity demanded of a good that results from the effect of a change in price of consumer purchasing power, holding all other factors constant.

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- Only for normal goods will the income effect results in consumers increasing the quantity of the good they demanded when the price falls
- If the good is an inferior good, then the income effect leads consumers to decrease the quantity of the good they demand. **Answer: B**

83. The substitution effect describes how
    A. an increase in income causes buyers to purchase different products
    B. the opportunities for minorities are purchase different products
    C. the additional satisfaction from each additional unit of the product consumed in a given time period decreases
    D. a price increase creates incentives for buyers to seek alternatives

***Explanation*:**
The substitution effect occurs when the price of a good falls, consumers will substitute it for other goods, which are now relatively more expensive. **Answer: D**

84. If the income effect is the only economic principle operating, when price increases the quantity
    A. demanded of a normal good will increase
    B. demanded of an inferior good will increase
    C. demanded of a complement will increase
    D. demanded of any good will decrease

***Explanation*:**
The income effect occurs when a fall in the price of a good increases consumer's real income, making them more able to purchase all goods, so the quantity demanded increases.

Answer: B

Supplementary Exercise on Unit 3

1. Suppose your marginal utility from consuming the 3rd slice of cake is zero, then your total utility from consuming cake is
    A. negative. B. maximized. C. increasing. D. decreasing.
2. Total utility is:
    A. the amount of satisfaction that a consumer derives by consuming a commodity.
    B. the utility gained through the change in the form of the commodity.
    C. the aggregate sum of satisfaction that an individual gains by consuming a given amount of goods and services.
    D. the utility gained through the product consumed at the right time.
3. With each additional increase in consumption of a good, marginal utility usually
    A. remains constant. B. decreases. C. increases. D. exceeds total utility.
4. Utility refers to the
    A. satisfaction derived from consuming a good or service

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B. net social benefit of a good or service
C. market value of a good or service
D. scarcity price of a good or service

5. The term "marginal" in economics means
A. the minimum unit C. unimportant
B. the maximum unit D. additional

6. The satisfaction or pleasure one gets from consuming a good or service is:
A. price. B. utility. C. consumption. D. preferences

7. Marginal utility is
A. the utility per unit associated with the last unit of a good consumed
B. the total utility associated with consuming a good
C. the usefulness of the last or next unit of a good consumed
D. the change in total utility associated with consuming an additional unit of a good

8. Which of the following statements concerning indifference curves is FALSE?
A. An indifference curve shows all combinations of goods that give the same amount of satisfaction.
B. The further away from the origin an indifference curve is, the greater the amount of satisfaction it represents
C. An indifference curve's diminishing slope shows that the more of a good a consumer has the greater amount of it she is willing to exchange for another good
D. An increase in real income will cause an indifference curve to shift outward.

9. If marginal utility is positive but decreasing, total utility is
A. decreasing B. negative. C. increasing. D. constant

10. According to the law of diminishing marginal utility, as quantity consumed increases
A. marginal utility increases
B. total utility decreases
C. marginal utility decreases
D. both marginal and total utility increase

11. At the utility-maximizing level of consumption,
A. The last dollar spent on each good produces the same with marginal utility.
B. Total expenditure equals total utility.
C. Marginal expenditure is the same for both goods in question.
D. Consumption is at competitive equilibrium

12. Suppose Cabbage is free, then to maximize total utility, a consumer should eat Cabbage until his/her marginal utility is:

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A. at a maximum
B. Zero
C. equals the price of tea
D. no longer diminishing

13. Assume MUx = 1,000 utils, Px = 50, MUy = 250 and Py = 20. This consumer
A. should buy less of X and less of Y.
B. should buy more of X and less of Y.
C. is in equilibrium.
D. should buy more of X and more of Y.

14. The marginal rate of substitution between food and shelter for a given point on an indifference curve:
A. is equal to the absolute value of the slope of the indifference curve at that point.
B. is equal to the rate at which the consumer is willing to exchange the two goods in the marketplace.
C. reflects the relative values the consumer attaches to the two good.
D. is described, in part, by each of the above statements.

15. The consumer maximizes personal satisfaction when
A. MUa/Pa = Pb/MUb
B. Pa/MUa = Pb/MUb
C. MUa/Pa = MUb/Pb
D. MUa/Pb = Pa/MUb

16. If total utility is decreasing, then marginal utility is
A. negative. B. positive. C. zero. D. increasing.

17. If the prices of both goods increase by the same percent, the budget line will
A. shift parallel to the left.
B. shift parallel to the right.
C. pivot about the x axis.
D. pivot about the y axis.

18. A utility contour (or indifference curve) shows all the alternative combinations of two consumption goods that
A. can be produced with a given set of resources and technology.
B. yields the same total of utility.
C. can be purchased with a given budget at given prices.
D. equate the marginal utilities of these goods and, therefore, make the consumer indifferent between them.

19. Consumer Choice or Utility Analysis is the analysis of
A. the formation of tastes and preferences.
B. the differences of tastes and preferences across individuals.
C. consumption decisions of people based on utility maximization.
D. how consumers determine the utility they receive from consumption.

20. When the marginal utility per dollar of good x exceeds the marginal utility per dollar of good y,
A. the consumer should consume more x.
B. the consumer is consuming too much x.

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C. good y must have a negative marginal utility.

D. the consumer is in an optimal situation if the price of x exceeds the price of y.

21. Suppose a consumer has an income of Birr 16, the price of A is Birr 2, and the price of B is Birr 1. Which combination is on the consumer's budget line?

A. 4A and 9B B. 6A and 6B C. 8A and 8B D. 5A and 6B

22. If a consumer has an income of Birr 200, the price of X is Birr 5, and the price of Y is Birr 10, the maximum quantity of X the consumer is able to purchase is:

A. 5. B. 10. C. 20. D. 40.

23. A woman has Birr 500 a week to spend on clothing and food. The price of clothing is Birr 25 and the price of food is Birr 10. She spends her entire income when she purchases_____ units of clothing and_____ units of food.

A. 10; 10 B. 25; 5 C. 12; 20 D. 16; 8

24. Explain the law of diminishing marginal utility
25. List the properties of indifference curves
26. An individual marginal utilities for commodities are given as follows:

MUX= 50 and MUY = 100

What is marginal rate of substitution commodity A for commodity B

27. Consumer X has Birr 20 per week to spend on any combination of pineapples and green tea. The price of a pineapple is Birr 4 and the price of a bottle of green tea is Birr 2. The table below shows the consumer's utility values. What is the consumer utility maximizing bundle?

Quantity of Pineapples	Total Utility	Quantity of Green Tea	Total Utility
1	32	1	28
2	52	2	46
3	64	3	54
4	68	4	60
5	70	5	64
6	71	6	66
7	71	7	67

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Answer to Supplementary Exercise on Unit 3

1. B 2. C 3. B 4. A 5. D 6. B 7. D 8. C 9. C 10. C 11. A
12. B 13. B 14. B 15. C 16. A 17. A 18. B 19. C 20. A 21. D 22. D
23. C

24. Law of diminishing marginal utility states that as the amount consumed of a commodity increases the utility derived by the consumer from the additional unit goes on decreasing.

25. The properties of indifference curve are:
- An indifference curve slopes down from left to right
- An indifference curve is concave to the origin
- Two indifference curve do not intersect each other
- Higher indifference curve gives higher satisfaction

26. MRS AB = 2

Pineapple	TU	MU	MUA/PA	Green tea	TU	MU	MUT/PT
1	32	-	-	1	28	-	-
2	52	20	5	2	46	18	9
3	64	12	3	3	54	8	4
4	68	4	1	4	60	6	3
5	70	2	0.5	5	64	4	2
6	71	1	0.25	6	66	2	1
7	71	0	-	7	67	1	0.5

MUA/PA = MUT/PT

QAXPA+QTXT = 20

The maximum utility is obtained at 3 units of Pine apple and 4 units of Green Tea.

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UNIT FOUR

Theory of Production and Cost

4.1 Theory of production:

Production theory is the study of **production**, or the economic process of converting inputs into outputs. Production uses resources to create a good or service that is suitable for use, gift-giving in a gift economy, or exchange in a market economy. This can include manufacturing, storing, shipping, and packaging.

Production refers to the output of goods and services produced by businesses within a market.

This production creates the supply that allows our needs and wants to be satisfied.

A production process can be defined as any activity that increases the similarity between the pattern of demand for goods and services, and the quantity, form, shape, size, length and distribution of these goods and services available to the market place.

4.1.1 Production Inputs:

The inputs or **resources** used in the production process are called *factors of production.*

Fixed Input: A fixed input is an input used in production and under the control of the producer that does not change during the time period of analysis (the short run). Examples include major pieces of equipment, suitable factory space, and key managerial personnel.

Variable Input: A variable input is an input used in production and under the control of the producer that does change during the time period of analysis (the short run). Examples include electrical power consumption, transportation services, and most raw material inputs.

4.1.2 Period of production:

In the theory of the firm the distinction between short run and long run is *not* necessarily based on the length of time. It is rather based on the degree of the variability of inputs. There are two types of production period:

Short run production period:

The short run is a period of time when there is **at least one fixed factor input**. This is usually the capital input such as plant and machinery and the

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stock of buildings and technology. In the short run, the output of a business expands when more variable factors of production (e.g. labor, raw materials and components) are employed.

Long run production period:

In the long run, **all of the factors of production can change** giving a business the opportunity to increase the **scale of its operations.** For example a business may grow by adding extra labor and capital to the production process and introducing new technology into their operations.

4.1.3 Production function:

A **production function** is the mechanism for combining production resources, with existing technology, into finished goods and services. In other words, a production function takes inputs and creates output. In a production function that uses only labour (L) and capital (K):

A **production function** relates physical output of a production process to physical inputs or factors of production

A production function can be expressed in a functional form as:

$$Q = f(x_1, x_2, x_3, \ldots x_n,$$

Where:

Q = Quantity of output

$X_1, X_2, X_3, \ldots, X_n$ = Quantities of factor inputs (such as capital, labor, land or raw materials).

Productions function with one variable input:

Total product: The total product (or total physical product) of a variable factor of production identifies what outputs are possible using various levels of the variable input. It is the maximum quantity hat can be produced from a given combination of inputs.

Average Product: The average physical product is the total production divided by the number of units of variable input employed. It is the output of each unit of input.

$$AP_L = Q/L$$

Marginal Product: is the additional output that is generated by an additional worker. It is the change in total product when one of the inputs is changed. Marginal product =Change in output/change in input

$$MP_L = dq/dL$$

Relationship between total product and marginal product:

- When TP increases at increasing rate, MP increases.
- When TP increases at decreasing rate, MP decreases
- When TP reaches maximum, MP is zero
- When TP decreases, MP is negative

Relationship between average product and marginal product:

MPL > APL AP is increasing

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MPL < APL AP is decreasing
MPL = APL AP is flat

Illustrative Examples

1. Production function represents:
   A. The maximum amounts of inputs required to product a given amount of output
   B. The maximum amount of output that can be obtained from a given amount of input
   C. The minimum amount of inputs required to produce a given amount of output
   D. Both B and C are correct.

***Explanation*:**
- Production function is a model that relates possible levels of physical outputs to various sets of inputs
- it is the maximum amount of output that can be obtained from a given amount of input
- it is the minimum amount of inputs required to produce a given amount of output

Answer: D

2. Production in the short run is:
   A. The period of time during which at least one factor of production is fixed
   B. The period of time during which all factors of production are variable.
   C. The period of time during which all factors of production are fixed.
   D. Less than one year.

***Explanation*:** Short run is a period in which technology is constant, at least one input is fixed and at least one input is variable. **Answer: A**

3. Production in the long run is:
   A. The period of time during which at least one factor of production is fixed.
   B. The period of time during which all factors of production are variable.
   C. The period of time during which all factors of production are fixed.
   D. Greater than one year.

***Explanation*:** A period in which all inputs are variable but technology is constant **Answer: B**

4. The production function:
   A. Is only applicable in the short run
   B. Cannot be empirically estimated

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C. Summarize the relationship between output and factors of production

D. Summarize the least-cost combination of inputs to produce a give output.

***Explanation*:**

- Production function indicates the maximum amount of output that can be produced with the help of each possible combination of inputs
- For example: Q = f (Labour, Kapital, Land, technology, where Q is output, labour, capital, land and technology are inputs.

Answer: C

5. The firm operates in the, but plans in the .

A. long-run; short-run
B. long-run; long-run
C. short-run; long-run
D. short-run; short-run

***Explanation*:** The firm operates in the short run but plans in the long run

Answer: C

6. Which of the following explains production in the long run when there are two variable inputs?

A. The law of variable proportions
B. The law of production possibilities
C. The law of returns to scale
D. The law of diminishing returns

***Explanation*:**

- The law of variable proportions is explain production in the short run
- The law of returns to scale is the property that applies only in the long run

Answer: C

7. When the marginal product of labour is equal to the average product of labour,

A. marginal product of labour is at its maximum.
B. marginal cost of production is at its minimum.
C. marginal cost is equal to minimum average variable cost.
D. average total cost is at its minimum.

***Explanation*:**

- When marginal product equals average product, average product reaches its maximum
- When marginal product equals to average product, the marginal cost is equal to average variable cost

Answer: C

8. The additional product that results from the addition of one more worker is called the:

A. Total product of labour
B. Average product of labour
C. Marginal product of labour
D. Zero product of labour

***Explanation*:** The marginal product of a factor is the addition to the total production by the employment of an extra unit of a factor

Answer: C

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The table below shows the product schedules a producer of cement blocks. Assume that labour is the variable input and the block machine is the fixed input. Complete the marginal product column of the table. On the basis of this information, answer the next question.

Labour (workers/day)	Total Product (blocks/day)	Marginal product (blocks/day)
0	20	
1	40	
2	80	
3	120	
4	150	
5	160	
6	150	

9. The firm realizes increasing marginal returns for the first:

A. 2 workers hired
B. 3 workers hired
C. 4 workers hired
D. 5 workers hired

Solution:

Labor	Total product	Marginal product
1	0	-
2	20	20
3	80	60
4	120	40
5	150	30
6	160	10
7	150	-10

The firm realizes increasing marginal returns after hiring the 2nd worker

Answer: A

10. If the marginal product of an input factor is falling, then

A. average fixed cost is constant.
B. marginal cost is falling.
C. average total cost is constant
D. marginal cost is rising

Explanation:

- Marginal product and marginal cost are inversely related
- When marginal product of an input factor is falling, then the marginal cost is rising.

Answer: D

11. Which of the following is correct about marginal and average products?

A. When the marginal product is increasing, the average product must be increasing.

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B. When the marginal product exceeds the average product, the average product must be increasing.
C. When the average product is increasing, the marginal product must be decreasing.
D. When the marginal product is decreasing, the average product must be decreasing.

***Explanation*:**

- When MP > AP, then AP increases
- When MP < AP, then AP decreases
- When MP = AP, then AP reaches its maximum

Answer: B

12. When the total product of a resource is at a maximum then:
A. average product is equal to marginal product
B. average product is equal to zero
C. marginal product is equal to zero
D. average product is at its maximum

***Explanation*:**

- When total product increases at an increasing rate, marginal product increases.
- When total product increases at a diminishing rate, marginal product declines.
- When total product reaches its maximum, marginal product becomes zero.
- When total product begins to decline, marginal product becomes negative.

Answer: C

13. Increasing marginal returns occur
A. Through the entire range of production
B. As output increases at high levels of production
C. As output increases at low levels of production
D. Whenever total product increases at an increasing rate

***Explanation*:**

- When total product increases at increasing rate, then marginal product increases
- Increasing marginal returns occur when total product increases at an increasing rate

Answer: D

14. If total product of labour is rising at an increasing rate,
A. marginal product of labour is rising.
B. marginal product of labour is at its minimum.
C. marginal product of labour is at its maximum.
D. average product of labour is at its minimum.

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***Explanation*:**

- MPL tells you how TPL is changing when more labour is hired. If more labour is increasing
- TPL at a faster and faster rate, MPL is rising.

Answer: A

4.1.4 The law of diminishing marginal returns:

The **law of diminishing marginal returns** states that as successive amounts of the variable input, i.e., labor, are added to a fixed amount of other resources, i.e., capital, in the production process the marginal contribution of the additional variable resource will eventually decline.

The **law of diminishing returns** occurs because:

- Unless all inputs are perfectly and infinitely substitutable, as we increase the amount of one input, while keeping other inputs constant, at some point the productive effectiveness of that input starts to decline.
- This means that resources used in producing one type of product are not necessarily as efficient (or productive) when switched to the production of another good or service.
- For example, workers employed in producing coble stone for use in the construction industry may not be as efficient if they have to be re-employed in producing cement or ceramics industry.

Likewise many items of capital equipment are specific to one type of production.

Note:

- ***Increasing marginal returns occurs when MPL increase as L increases.***
- ***Diminishing marginal returns occurs when MPL decreases as L increases.***
- ***Negative marginal returns occurs when MPL becomes negative as L increases***

Illustrative Examples

15. Diminishing marginal returns occurs when
    A .All inputs are increased and output increases by a greater proportion
    B. All inputs are increased and output increases by a smaller proportion
    C. A variable input is increased and its marginal product rises
    D. A variable input is increased and its marginal product falls

***Explanation*:** Diminishing marginal returns occur when additional quantities of the variable input will yield diminishing marginal returns to total product.

Answer: D

16. Diminishing marginal product occurs
    A. immediately after the first worker is hired.
    B. after the marginal cost curve crosses the average variable cost curve.

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C. at different times for different firms.
D. when average variable cost begins to fall.

Explanation:

- Diminishing marginal product occurs immediately after the marginal cost curve crosses the average total variable cost curve.
- After the intersection point of MC and AVC, MC rises and MP falls because MC and MP are inversely related. **Answer: B**

17. The law of diminishing returns is a law while the law of returns to scale is a law.

A. short run, short run
B. short run, long run
C. long run, short run
D. long run, long run

Explanation: The law of diminishing returns is a short run phenomena while returns to scale is a long run phenomenon **Answer: B**

18. Which of the following conditions must exist for diminishing returns to set in?

A. All production inputs must proportionally increase in quantity
B. Increase in quantity of some inputs must come with fixed quantity of at least one other input
C. The skill and knowhow of workers must decline
D. The quantity and quality of all inputs must diminish

Explanation: The law of diminishing returns state that if more and more units of a variable input are applied to a given quantity of fixed inputs, the total output increases at diminishing rate. **Answer: B**

19. The Law of Diminishing Marginal Returns is responsible for

A. AVC that first rises, but eventually falls, as output increases.
B. AFC that first rises, but eventually falls, as output increases.
C. MP that first falls, but eventually rises, as output increases.
D. MC that first falls, but eventually rises, as output increases

Explanation:

- The Law of Diminishing Marginal Returns says that MPL eventually falls as you add more labor to a fixed plant.
- Remember that MC = w/MPL. So when MPL is initially rising, MC is falling
- Eventually when MPL is falling, MC is rising.
- AFC continues to fall, but AVC and ATC eventually rise. **Answer: D**

20. What Are the Advantages of the Law of Diminishing Returns?

Explanation:

- The Law of Diminishing Returns is a generalization of an observation from economics that increasing one input into an economic process while holding the other factors constant, will produce smaller and smaller improvements in the output.

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- The advantages of applying the law are that it will help avoid a misapplication of resources.
- According to the law of diminishing returns, you may increase an input to obtain an improved result but the returns will diminish until it is more effective to increase a different input

4.1.5 Stage of production:

Economists identify *three stages of production* on the basis of the *marginal* and *average* returns to the variable input, i.e., labor (*L*)

- *Stage I* : *The marginal product* of the variable input (labor) is higher than its *average product*, i.e., *MPL > APL*
- *Stage II* : *The marginal product* of the variable input (labor) falls below its *average product*, i.e., in Stage II, *MPL < APL*, but both remaining greater than zero.
- *Stage III* : *The marginal product* of the variable input (labor) turns *negative*, while *average product* remains greater than zero.

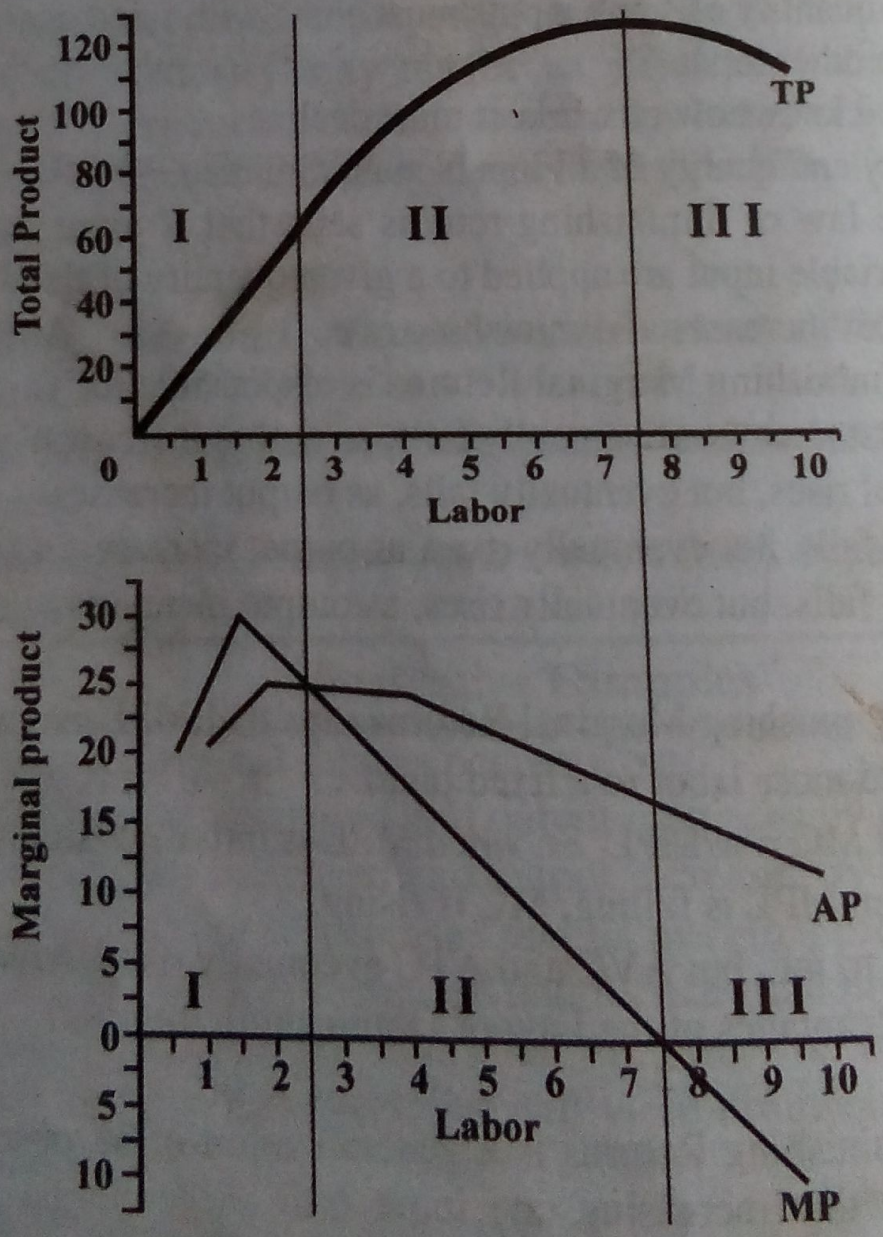

Figure 4.1: stage of production.

In figure 4.1:

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- The top panel contains the total product curve (TP). It generally rises, reaches a peak, and then falls.
- The bottom panel contains the marginal product curve (MP) and the average product curve (AP).
- Both curves rise a bit for small quantities of the variable input labor, then decline.
- Marginal product eventually turns negative, but average product remains positive.

Stage I

Short-run production Stage I arise due to increasing average product. As more of the variable input is added to the fixed input, the marginal product of the variable input increases. Most importantly, marginal product is greater than average product, which causes average product to increase In stage one:

- The total product curve has a positive slope.
- Marginal product is greater than average product.
- Marginal product initially increases then decreases until it is equal to average product at the end of Stage I.
- Average product is positive and the average product curve has a positive slope.

Stage II

In Stage II:

- The total product curve has a decreasing positive slope. In other words, the slope becomes flatter with each additional unit of variable input.
- Marginal product is positive and the marginal product curve has a negative slope. The marginal product curve intersects the horizontal quantity axis at the end of Stage II.
- Average product is positive and the average product curve has a negative slope. The average product curve is at it's a peak at the onset of Stage II. At this peak, average product is equal to marginal product.

Stage III

- The total product curve has a negative slope. It has passed its peak and is heading down.
- Marginal product is negative and the marginal product curve has a negative slope. The marginal product curve has intersected the horizontal axis and is moving down.
- Average product remains positive but the average product curve has a negative slope.

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Economic Production

- In stage I the variable input is increasingly more productive, firms employ as many as they can, as quickly as they can.
- In Stage III, with negative marginal returns, is not particularly attractive to firms. The cost of production is greater due to the employment of the variable input.
- Stage II, with decreasing but positive marginal returns, provides a range of production that is suitable to most every firm.

As a matter of fact, because Stage II tends to be the choice of firms for short-run production, it is often referred to as the "economic region." Firms quickly move from Stage I to Stage II, and do all they can to avoid moving into Stage III. Firms can comfortably, and profitably, produce forever and ever in Stage II.

Illustrative Examples

21. Consider the production function $Q = f(K, L)$, where K is capital and L is labor. If $MPL < 0$, then the firm must be operating in:
    A. Stage I of production.
    B. Stage II of production.
    C. Stage III of production
    D. Stage IV of production.

***Explanation*:** Stage III of production is the range of product where the marginal product of labor is negative $MPL < 0$ **Answer: C**

22. In Stage II of production:
    A. *TPL* must be increasing at an increasing rate.
    B. *MPL* must be greater than *APL*.
    C. *APL* must be greater than *MPL*.
    D. *APL* is increasing.

***Explanation*:**

- Stage II of production is the range of output where the average product of labor is declining and the marginal product of labor is positive.
- In other words, stage II of production begins where APL is maximized, and ends with MPL = 0.
- In stage II AP must be greater than MP

Answer: C

23. Stage I of production ends where:
    A. *APL = MPL*.
    B. *APL* is maximized.
    C. *APL = 0*.
    D. Both A and B are correct

***Explanation*:**

- stage I of production is the range of labor usage where the average product of labor (APL) is increasing.
- Over this range of output, the marginal product of labor (MPL) is greater than the average product of labor.

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- Stage I ends and stage II begins where the average product of labor is maximized, i.e., APL = MPL.

Answer: D

24. Stage II of production ends where:
    A. *MPL* is maximized.
    B. *MPL* = *0*.
    C. *TPL* is maximized
    D. Both B and C are correct.

***Explanation*:**

- stage II of production begins where APL is maximized, and ends with MPL = 0.
- At the end of stage two total product is maximum

Answer: D

25. In the second stage of the short run production function in which labour is the only variable input, which of the following is true about the relationship between average product (APL) and marginal product (MPL)?
    A. APL is greater than MPL and MPL is positive
    B. APL is greater than MPL and MPL is negative
    C. APL is less than MPL and MPL is positive
    D. APL is equal to MPL and APL is at its maximum

***Explanation*:** *In the second stage of short run production function, the marginal product* of the variable input (labor) falls below its *average product,* i.e., in Stage II, *MPL < APL*, but both remaining greater than zero.

Answer: A

4.1.6 Production functions with two variable inputs:

Isoquants: As stated in the book, an isoquant is a curve that shows all the combinations of labor and capital that can produce a given level of output.

Properties of Iso-quants:

- **Isoquants have a negative slope**. In order to maintain one level of output, when the amount of one factor is increased, that of the other is decreased. At each point on the iso-quant curve, we get a combination of two factors, which give the same level of output.
- **Isoquants are convex to the origin**: The convexity of the isoquant curve suggests that MRTS is diminishing, meaning, when quantity of one factor is increased, the less of another factor will be given up, keeping the output constant.
- **Iso-quants do not intersect:** Each Isoquant represents a specific quantum of output. If two Isoquants intersect each other, it would lead to a logical contradiction as Isoquant representing a smaller quantity cannot be on a line representing a larger quantity.

Illustrative Examples

26. An isoquant:

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A. Reflects the various combinations of output that can be produced from a given amount of resources.
B. Reflects the various combinations of two inputs that will produce a given amount of output.
C. Reflects the various combinations of two inputs that cost a given amount.
D. Reflects the various combinations of two products that yield a given amount of utility.

***Explanation*:**
- An isoquant or equal product curve represents the various combinations of inputs that give the firm the same level of output. **Answer: B**

27. If an isoquant is convex to the origin (bowed inward)
A. More and more of one input is required to substitute for each successive unit reduction in the other input.
B. Less and less of one input is required to substitute for each successive unit reduction in the other input
C. The cost of producing a given amount of output will be the same regardless of the combination of the two inputs that are employed.
D. The producer will be indifferent between various input combinations because they will all yield the same profit.

***Explanation*:**
- The property of convexity implies that the slope of the isoquant diminishes from left to right along the curve.
- Convexity of isoquant is the result of diminishing marginal rate of technical substitution.
- Marginal rate of technical substitution diminishes as more and more labour is substituted for capital. **Answer: A**

28. An isoquant represents combinations of inputs that
A. Produce increasing amounts of output
B. Produce decreasing amounts of output
C. Produce the same level of output
D. Create wealth

***Explanation*:** An isoquant represents all those factor combinations which are capable of producing the same level of output **Answer: C**

29. An isoquant that is:
A. Further from the origin represents less output
B. Further from the origin represents greater output
C. Flatter represents less output
D. Steeper represents greater output

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Explanation:

- Higher indifference curve represents a higher level of satisfaction than a lower one.
- Further from the origin represents greater output **Answer: B**

30. If an isoquant is linear, what does this imply about the nature of the inputs in the production process?

A. They are prefect substitutes C. They are unrelated
B. They are perfect complements D. They exhibit diminishing returns

Explanation:

- When the two factors are perfect substitutes of each other, then each of them can be used equally well in the place of the other
- The isoquant of perfect substitutes are straight line **Answer: A**

31. If an isoquant is L-shaped, what does this imply about the nature of the inputs in the production process?

A. They are perfect substitutes C. They are unrelated
B. They are perfect complements D. They exhibit diminishing returns

Explanation:

- The perfect complementary factors are those which are jointly used for production in a fixed proportion.
- The isoquant of perfect complements are L – shaped **Answer: B**

32. Isoquants usually slope downward from left to right because

A. Marginal products of L and K are usually positive
B. Marginal products of L and K are always positive
C. Marginal products of L and K are eventually decrease
D. Marginal products of L and K are always together

Explanation:

- The negative slope of the isoquant implies that if one of the inputs is reduced, the other inputs have to be so increased that the total output remains unaffected.
- The downward slopping property of isoquants follows from the assumption that the marginal physical products of factors are positive, that is the use of additional units of factors yield positive increments in outputs. **Answer: B**

4.1.7 Marginal rate of technical substitution:

The Marginal Rate of Technical Substitution (MRTS) or Technical Rate of Substitution (TRS) is the amount by which the quantity of one input has to be reduced when one extra unit of another input is used so that output remains constant.

$$\mathbf{MRTS_{L,K}} = \frac{\Delta K}{\Delta L} = \frac{MPL}{MPK}$$

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Where, MPL and MPK are the marginal products labor and capital, respectively, and MRTS is Marginal Rate of Technical Substitution of the input L for K.

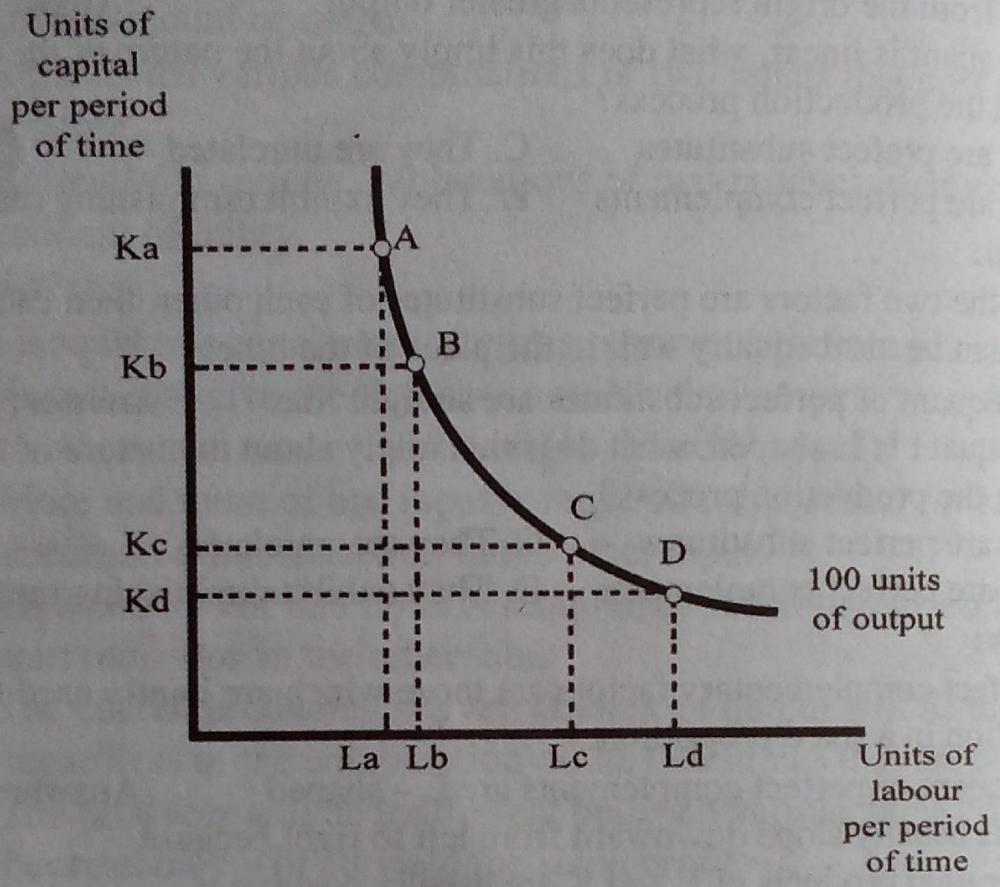

Figure 4.2: marginal rate of technical substitution.

Along an isoquant, the MRTS shows the rate at which one input (e.g. capital or labor) may be substituted for another, while maintaining the same level of output(100 units of output)

The marginal rate of substitution of labour for capital gives the amount of capital that can be replaced by one unit of labour while keeping output unchanged.

To move from point A to point B in the diagram, the amount of capital is reduced from Ka to Kb while the amount of labour is increased only from La to Lb. To move from point C to point D, the amount of capital is reduced from Kc to Kd while the amount of labour is increased from Lc to Ld. The marginal rate of technical substitution of labour for capital is equivalent to the absolute slope of the isoquant at that point (change in capital divided by change in labour). It is equal to 0 where the isoquant becomes horizontal, and equal to infinity where it becomes vertical.

Illustrative Examples

33. The marginal rate of technical substitution is the rate at which
    A. Outputs can be substitution for one another
    B. One input can be substitute for another with varying output

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C. New substitutes can be used for current inputs

D. One input substitutes for another without changing output

***Explanation*:** Marginal rate of technical substitution is the rate at which one factor can be substituted for another while holding the level of output constant

Answer: D

34. A firm reports that marginal product of capital is 6 and marginal rate of technical substitution of capital for labor is 3. What is the marginal product of labor?

***Solution*:**

Given: MPK = 6, MRTSKL = 3, MPL =?

MRTSKL= MPK/MPL

3 = 6/MPL, MPL = 6/3

MPL = 2

35. The marginal rate of technical substitution for L and K is defined by

A.MRTSLK =MPL/MPK C. MRTSLK=MPL xMPK

B. MRTSLK =MPK/MPL D. MRTSLK=MPKxMPL

***Explanation*:**

MRTSLK =MPL/MPK

MRTSKL = MPK /MPL

Answer: A

36. If the price of labor is Birr 4 and the price of capital is Birr 16, what is the marginal rate of technical substitution at the optimal input choice?

***Solution*:**

Given: PL = 4, PK = 16

MRTS = MPL/MPK = PL/PK

MRTS = PL/PK

MRS = 4/16, = ¼= 0.25

37. The slope of the isoquant is the marginal rate of technical substitution

A. *MPK / MPL* B. $-\Delta K/\Delta L$. C. both *a* and *c* D. all of the above

***Explanation*:**

- The slope of an isoquant at each point equals the change in capital divided by the change in labor, holding output constant. This slope is called the marginal rate of substitution.
- Isoquants are negatively sloped.
- MRTSLK = - K/ L

Answer: B

4.1.8 Isocost Lines

Isocost line is a line representing all possible combinations of inputs that can be purchased for a given total cost.

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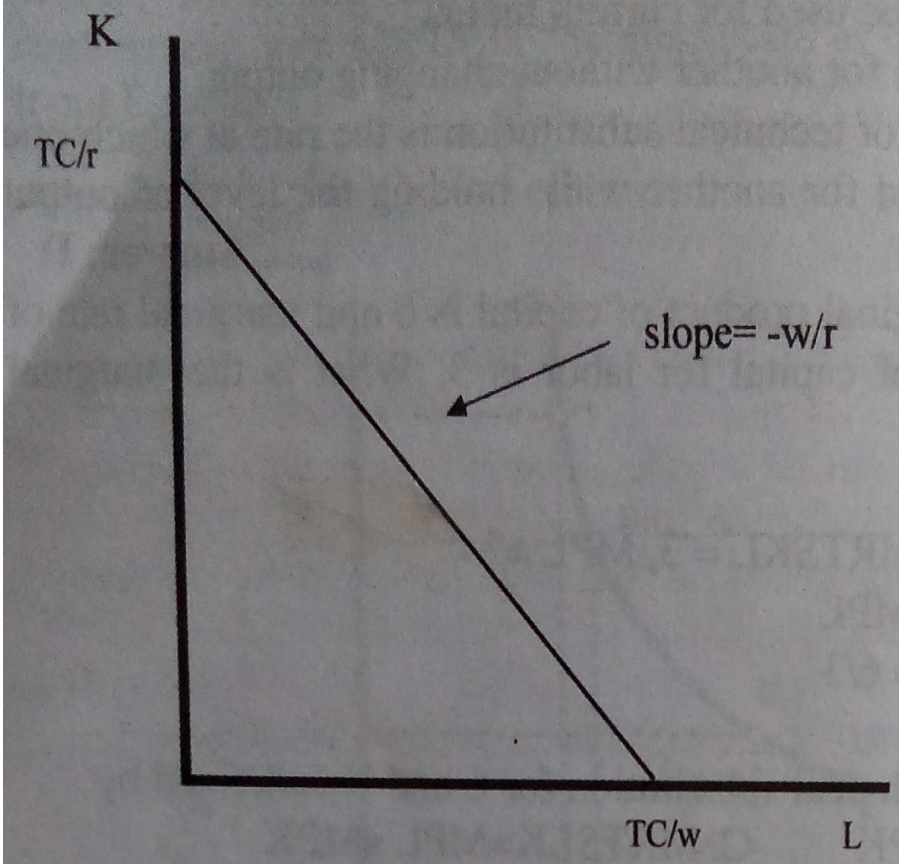

- Costs increase as the isocost line shifts out.
- An isocost line is similar to the budget constraint from consumer theory.

Illustrative Examples

38. Assume that a firm spends Birr 500 on two inputs, labor (graphed on the horizontal axis) and capital (graphed on the vertical axis). If the wage rate is Birr 20 per hour and the rental cost of capital is Birr 25 per hour, the slope of the isocost curve will be

***Solution*:**

- The slope of the isocost line is equal to the ratio of price ratio of price of labour and price of capital.
- Slope of isocost curve = price of labour / price of capital

Slope = -PL / PK= - w /r

= -20/ 25= -4/5

39. At the optimum combination of two inputs,
    A. the slopes of the isoquant and isocost curves are equal.
    B. costs are minimized for the production of a given output.
    C. the marginal rate of technical substitution equals the ratio of input price
    D. all of the above

***Explanation*:**

- The optimum combinations of two inputs occur at a point of tangency.
- The tangency point shows that optimization in production is reached when factor prices and marginal product are proportional, with equalized marginal product.

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- MRTSLK = w/r, where MRTSLK is the slope of isoquants and w/r is the slope of isocost line.

Answer: D

40. Suppose that the price of labor (*PL*) is Birr 10 and the price of capital (*PK*) is Birr 20. What is the equation of the isocost line corresponding to a total cost of Birr 100?

A. PL + 20PK
B. 100 = 10L + 20K
C. 100 = 30(L+K)
D. 100 + 30 (*PL* + *PK*)

***Explanation*:** The equation of the isocost line is the total cost incurred on the factors of production for producing a commodity equal to the sum of the payments made to labor and capital.

C = w.L + r.K, where C is total cost, w is wage rate for labor and r is price of capital.

100 = 10L + 20K

Answer: B

41. If capital is on the vertical axis and labor is on the horizontal axis, the slope of an isocost line is given by

A. - PL/Pk B. - Pk/PL C. - Pk x PL D. - PL x Pk

***Explanation*:** The slope of the isocost line is equal to the ratio of price ratio of price of labour and price of capital

Answer: A

42. Assume that the marginal product of labor is 2, the marginal product of capital is 4, and the wage rate is Birr 3, the rental price of capital is Birr 6, and the price of output is Birr 1.50. The firm

A. Operates at the point of maximum efficiency/least cost
B. Could increase efficiency by hiring more labor and less capital
C. Could increase efficiency by hiring more capital and less labor
D. All of the above are correct

***Solution*:**

Given: MPL = 2, MPK = 4 , w = 3, r = 6

Efficiency condition for factor use requires that the following conditions should be fulfilled

MPL/w = MPK/ r

2/3 = 4/6 = 2/3 = 2/ 3

So the firm operates at the point of maximum efficiency/least cost

Answer: A

43. A tangency point between an isoquant and an isocost line identifies

A. the least costly combination of inputs required to produce various levels of outputs.
B. the various levels of output that can be produced using a given level of inputs.
C. the various combinations of inputs that can be used to produce a given level of output

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D. the least costly combination of inputs required to produce a given level of output

Explanation:

- The necessary condition of minimum cost is satisfied at the point of tangency between the isoquant and the isocost.
- At the point of tangency, the marginal exchange ratio of inputs is equal to the ratio of their marginal productivity, i.e, at this point PL/PK = MPL/MPK
- At tangency point, the least costly combination of inputs required to produce a given level of output

Answer: D

44. Give an intuitive explanation that the relationship MPa/Pa = MPb/Pb = ... = Mpn/Pn must hold if the firm is to obtain maximum production from its budget for productive inputs. Then show algebraically that this is the same as saying that the slope of the isoquant must be equal to the slope of the isocost line if the firm is to obtain maximum production from its budget for productive inputs.

***Explanation*:** When one input has a higher MP/P, it's return, being very high, indicates it is being underused. You use less of other inputs so you can afford more of this input, thereby driving its productivity down, until all the marginal productivities (per last dollar spent) are equal.

Note; MRTSxy = Px/Py where the isoquant and isocost lines are tangent. This can be rewritten as MRTSxy = change in y/change in x = MPx/MPy = Px/Py Rewriting the last two terms, we get MPx/Px = Mpy/Py.

4.1.9 Law of Returns to Scale:

The law of returns to scale describes the relationship between variable inputs and output when all the inputs or factors are increased in the same proportion. The law of returns to scale analysis the effects of scale on the level of output

Increasing Returns to Scale:

If the output of a firm increases more than in proportion to an equal percentage increase in all inputs, the production is said to exhibit increasing returns to scale.

For example, if the amount of inputs are doubled and the output increases by more than double, it is said to be an increasing returns to scale.

The causes of economies of scale are:

- labor specialization
- managerial specialization
- more efficient capital
- ability to profitably use by-products

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Constant Returns to Scale:
When all inputs are increased by a certain percentage, the output increases by the same percentage, the production function is said to exhibit constant returns to scale.
For example, if a firm doubles inputs, it doubles output.
Diminishing Returns to Scale:
The term 'diminishing' returns to scale refers to scale where output increases in a smaller proportion than the increase in all inputs.
For example, if a firm increases inputs by 100% but the output decreases by less than 100%, the firm is said to exhibit decreasing returns to scale.
The cause of diseconomies scale is to the fact that management loses control of the firm beyond some size.
Economies of scale help determine the shape of the long run AC curve

- MC < AC (AC falling) => economies of scale
- MC > AC (AC rising) => diseconomies of scale

Illustrative Examples

45. A firm's short-run average cost curve is U-shaped. Which of these conclusions can be reached regarding the firm's returns to scale?
    A. The firm experiences increasing returns to scale.
    B. The firm experiences increasing, constant, and decreasing returns in that order.
    C. The firm experiences first decreasing, then increasing returns to scale.
    D. The short-run average cost curve reveals nothing regarding returns to scale.

***Explanation*:**

- Returns to scale is a property of production that indicates the relationship between proportionate change in all inputs and the resulting change in total product.
- It is a property that applies only in the long run **Answer: D**

46. When a firm is experiencing increasing returns to scale, what is happening to the long run total cost function and what is happening to the long run average cost function?

***Explanation*:** The long run total cost function is increasing at a decreasing rate, while the long run average cost function is decreasing.

47. Which of the following **is not** a factor which results in economies of scale?
    A. Bureaucracy in running a large organization
    B. More efficient utilization of the firm's by products
    C. Greater specialization and division of labor in the management of the firm

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D. Utilization of more efficient equipment

Explanation:

Reasons for operation of increasing returns to scale are:

- Greater division of labor and specialization
- Use of more productive specialized machinery and equipment
- Efficient utilization of the firm's by- products

Answer: A

48. Which of the following statements is entirely correct?
    A. Economies of scale are a short run concept, while diminishing returns is a long run concept.
    B. Both economies of scale and diminishing returns are short run concepts
    C. Both economies of scale and diminishing returns are long run concepts.
    D. Economies of scale is a long run concept, while diminishing returns is a short run concept

Explanation:

- Law of diminishing returns is a short run concept
- Returns to scale or economies of scale is a long run concept

Answer: D

49. When economies of scale exist, a firm's long-run average cost
    A. decreases as output increases
    B. slopes upward
    C. increases as output increases
    D. remains constant as output increases

Explanation:

- Economies of scale occur when a firm's long run average cost decreases
- Diseconomies of scale occur when long run average cost increases
- Economies of scale occur when you can double output for less than double the cost.
- Diseconomies of scale when you double output and costs more than double.

Answer: A

50. If we have increasing returns to scale and we increase the quantity of labor and capital used per unit of time by 10 percent, output will increase by
    A. Less than 10 %percent
    B. 10 percent
    C. Greater than 10 percent
    D. cannot be determined

Explanation:

- If the output of a firm increases more than in proportion to an equal percentage increase in all inputs, the production is said to exhibit increasing returns to scale.
- If inputs increases by 10%, output increases by more than 10%

Answer: C

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51. If we have decreasing returns to scale and we increase the quantity of labor and capital used per unit of time by 10 percent, output will increase by
A. Less than 10 percent
B. 10 percent
C. Greater than 10 percent
D. cannot be determined

Explanation:
- The term 'diminishing' returns to scale refers to scale where output increases in a smaller proportion than the increase in all inputs
- If labor and capital increases by 10 percent, output will increase by less than 10 percent.

Answer: A

52. If a firm triples all inputs, and output triples as well, the firm is subject to
A. constant returns to scale.
B. increasing returns to scale.
C. economies of scale.
D. both B and C

Explanation:
- When all inputs are increased by a certain percentage, the output increases by the same percentage, the production function is said to exhibit constant returns to scale.
- If inputs are triples, the output will triple.

Answer: A

53. In a certain production process, two inputs A and B yield an output of Q, whereas 3A and 3B yield an output of 4Q. This illustrates the principle of
A. Increasing marginal returns to variable input.
B. Diminishing returns to scale.
C. Increasing returns to scale
D. Increasing average returns to variable input.

Explanation:
- Q= F(inputs), inputs are A and B
- Inputs increases by 3 units and output increases by 4 units
- Output > inputs. It is increasing returns to scale.

Answer: C

54. A firm producing 500 units of output, with 20 units of capital and 100 units of labor, planned to expand its production to 750 units of output using 30 units of capital and 150 units of labor. What types of returns to scale is assumed in the planning?
A. Increasing B. Decreasing C. Constant D. Mixed

Explanation: Production is a function of labour and output

$Q = F[L,K]$

$500Q = 100L+20$

$750Q = 150L +30K$

$500/750 = 100/150 = 20/30$

$2/3=2/3=2/3$

Answer: C

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55. Returns to scale are generalizations about the patters of changes in output resulting from:
A. Technological changes
B. Changing management
C. Proportionate changes in the input use
D. Changing cost of production

***Explanation*:** Returns to scale are:
- a relationship between input and output
- the relationship between changes in production and the resulting change in output

Answer: C

4.2 Theory of cost:

4.2.1 Costs in the Short Run

Explicit Costs: Costs that involve a direct monetary outlay.
The firm actually pays out money for its inputs.
These payments include:
- *wages* and *salaries* for its workers,
- *rent* for its use of buildings and property,
- *interest* on any loans that were taken out to buy equipment
- Payments for raw materials.

Implicit Costs: The cost of inputs for which there is no direct money payment

Economic Costs: The sum of the firm's explicit costs and implicit costs.

Accounting Costs: Is the total of explicit costs that have been incurred in the past.

Sunk Costs: Costs that have already been incurred and cannot be recovered. Since sunk costs cannot be changed, they should not influence the firm's decision.

Non-sunk Costs: Costs that are incurred only if a particular decision is made.

Illustrative Examples

56. Which of the following statements is true regarding the differences between economic cost and accounting costs?
A. Accounting costs include all implicit and explicit costs.
B. Economic costs include implicit costs only.
C. Accountants consider only implicit costs when calculating costs.
D. Accounting costs include only explicit costs.

***Explanation*:**
- Accounting cost refers to the cost of purchased inputs only. It is the explicit cost of production only.
- Economic cost is the sum of explicit cost and implicit cost

Answer: D

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57. The costs of a firm that are paid directly in money are its:
A. implicit costs. B. explicit costs.
C. money opportunity costs. D. alternative costs.

Explanation: Explicit cost is the actual expenditure of the firm incurs to purchase inputs from the market

Answer: B

58. Explicit costs are and implicit costs are.
A. not measurable; measurable
B. paid in money; incurred when a firm gives up an alternate action
C. paid in money; paid in interest costs
D. used to maximize profits; used to minimize costs

Explanation:
- Explicit costs is the actual payment by firms to labor, capital, and other factors of production
- Implicit cost incurred by firms in producing a specific commodity consists of the amount that could be earned in the best alternative uses.

Answer: B

59. An example of an implicit cost of production would be
A. the cost of raw materials for producing bread in a bakery.
B. the cost of a delivery truck in a business that rarely makes deliveries.
C. the income an entrepreneur could have earned working for someone else
D. The cost of labor

Explanation: The cost of raw materials, the cost of a delivery truck and the cost of labor are explicit cost Implicit cost is the opportunity cost of the firms. The income an entrepreneur could have earned working for some else is an implicit cost.

Answer: C

60. Which of the following is most likely to be an implicit cost of production?
A. the "competitive rate" salary the owner of the business pays herself for services provided
B. the property taxes on a building owned by the firm
C. the rental income foregone because the business owns its building
D. the interest paid on outstanding loans of the business

Explanation:
- An implicit cost is incurred when an alternative is sacrificed.
- The rental income forgone because the business owns its building is an implicit cost.

Answer: C

61. The opportunity cost of a resource
A. includes both explicit and implicit costs
B. includes explicit costs only
C. includes implicit costs only

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***Explanation*:**

- Explicit costs are opportunity costs of a firm's resources that take the form of cash payments, for example, rent, mortgage, supplies, utilities, raw materials etc.
- Implicit costs are a firm's opportunity cost of using its own resources without corresponding cash payments.
- For example, when a firm uses money to pay for raw materials that it could charge interest free, the interest that the money could have made earning interest is an implicit cost.

Answer: A

62. When resources are owned by the firm, and no payment is made for their use, they

A. are considered to be free resources
B. have no alternative uses
C. represent implicit costs
D. represent explicit costs

***Explanation*:** Everything has an opportunity cost. What else could be done and how much money could be earned in regards to an owned resource? Whatever money could be earned in the best alternative use is the implicit cost of owning a resource. **Answer: C**

63. Which of the following is an implicit cost for the owner of a small store in your hometown?

A. The wage that is paid to the assistant manager.
B. The cost of purchasing canned goods from a wholesale food distributor.
C. The value placed on the owner's skills in an alternative career.
D. The cost of cooling the refrigerated meat display.

***Explanation*:** The opportunity cost of starting a small store is the salary given up in the next best alternative for the entrepreneur's skills.

Answer: C

4.2.2 Time element and cost:

Short Run: The period of time in which at least one of the firm's quantities cannot be changed

Long Run: The period of time that is long enough for the firm to vary the quantities of all its inputs as much as it desires.

Short- run cost of production:

Total Fixed Costs (TFC) are those costs that do not vary with changes in short-run output. They must be paid even when output is zero. These include:

- Rent paid on buildings and business rates charged by local authorities.
- The depreciation in the value of capital equipment due to age.
- Insurance charges.
- The costs of staff salaries e.g. for people employed on permanent contracts.

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- Interest charges on borrowed money.
- The costs of purchasing new capital equipment.
- Marketing and advertising costs.

2. **Total Variable Costs (TVC)** are those costs that change with the level of output. If output is zero, so are total variable costs. They include payment for
   - fuel, • Power, • transportation services, • the costs of raw materials,
   - labor costs
   - Operating expenses
   - other consumables and components used directly in the production process
3. **Total Cost (TC) is** the sum of total fixed and total variable costs at each level of output.

   TC = TVC + TFC

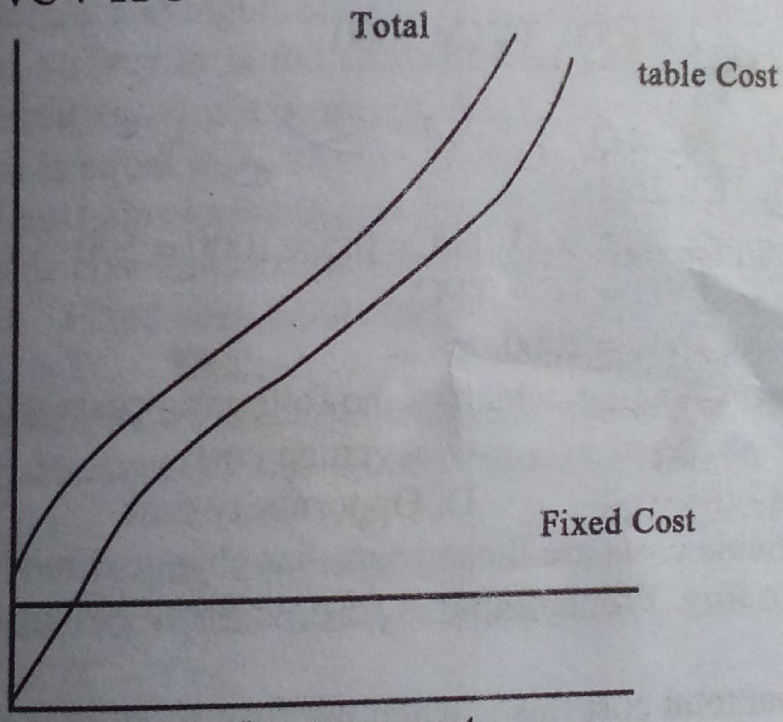

Figure 4.4: short run cost.

- ***The fixed cost curve is a horizontal line because they do not vary with quantity of output.***
- ***Variable cost has a positive slope because it varies with output.***
- ***The total cost curve has the same shape as the variable cost curve, but is above the variable cost curve by a distance equal to the amount of the fixed cost.***

Illustrative Examples

The average total cost to produce 100 cookies is Birr 0.25 per cookie. The marginal cost is constant at Birr 0.10 for all cookies produced.
64. Refer to the above Scenario, The total cost to produce 100 cookies is

Solution:

Average total cost = total cost / output

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ATC = TC / Q
TC= ATC x Q
TC= 0.25x100
TC= 25

65. How much is the total variable cost (TVC) of a firm, if the firm has an average total cost (ATC) of 2.5 Birr and an average fixed cost (AFC) of 0.5 Birr at an output level of 1000 units.

Solution:
Given: Q= 1000 units, ATC = 2.5 Birr, AFC = 0.5 Birr, TVC =?
ATC = AVC + AFC
AVC = ATC – AFC, AVC = 2.5-0.5= 2Birr
AVC = TVC/Q
TVC = AVC x Q = 2 x 1000, TVC= 2000
Or
ATC= TC/Q, TC= AC x Q
TC= 2.5 x 1000, TC= 2500
AFC = TFC/Q, TFC= AFC x Q, TFC= 0.5 x 1000 = 500
TC = TFC + TVC, TVC = TC – TFC
TVC= 2500 – 500, TVC = 2000

66. If a firm produces nothing, which of the following costs will be zero?
A. variable cost
B. Total cost
C. Average cost
D. Opportunity cost

Explanation: Variable costs are those costs that change directly with output level of the firm. When output is zero, variable cost also zero.

Answer: A

67. The amount that total cost raises when the firm produces one additional unit is
A. Marginal cost
B. Average cost
C. Fixed cost
D. Variable cost

Explanation: Marginal cost is the additional cost incurred to get one more unit of the output

Answer: A

68. If fixed cost at quantity (Q) = 100 is 130, then
A. fixed cost at Q = 0 is 0.
B. fixed cost at Q = 0 is less than 130.
C. fixed cost at Q = 200 is 260.
D. fixed cost at Q = 200 is 130.

Explanation: Fixed cost is the cost that does not vary as the firm changes its output level.
At Q= 0, fixed cost is 130, at Q = 200, fixed cost is 130. **Answer:** D

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69. Which of the following statements about a firm's short-run variable costs is correct?
A. They increase as the level of output decreases.
B. They typically include the cost of workers' wages.
C. They include the costs of plant and equipment.
D. They are always a greater expense than are fixed costs.

***Explanation*:** Variable cost is a cost that does change as the level of output changes. Variable cost increases when the firm expands the output level. Examples of variable costs for a business include:

- the costs of raw materials,
- wage of workers
- Operating expenses

Answer: B

70. The short run can be defined as any period of time:
A. Less than one year
B. In which some inputs are fixed
C. In which some inputs are variable
D. In which price is fixed

***Explanation*:** Short run is the period of time in which at least one of the firm's quantities cannot be changed

Answer: B

71. Fixed cost is equal to:
A. Total cost plus variable cost
B. Variable cost minus total cost
C. Total cost divided by output
D. Total cost minus variable cost

***Explanation*:** Fixed cost equals total cost minus total variable cost

$TFC = TC - TVC$

Answer: D

72. The Farley Farm, a dairy company, has total costs of Birr 15,000 and total variable costs of Birr 2,000. The Farley Farm's TOTAL FIXED COSTS are:

Given: $TC = 15{,}000$, $TVC = 2{,}000$

***Solution*:** Fixed cost equals total cost minus total variable cost

$TFC = TC - TVC$
$TFC = 15{,}000 - 2{,}000$
$TFC = 13{,}000$

73. In the short-run, total cost at zero output is
A. total fixed cost (TFC)
B. average total cost (ATC)
C. total variable cost (TVC)
D. average variable cost (AVC)

***Explanation*:** Total cost of production varies directly with the output level. Total cost at output level of zero is the value of total fixed cost.

Answer: A

74. Which of the following is true concerning short-run total costs?
A. total costs are minimized when average total costs are minimized
B. total costs are at a maximum when the average physical product of labor is at its maximum value
C. at zero output, total costs equal zero
D. total costs equal total variable costs plus total fixed costs

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***Explanation*:** Total cost is the sum of total fixed cost and total variable cost

Answer: D

75. A firm has a fixed production cost of Birr 5,000 and a constant marginal cost of production of Birr 500 per unit produced.

A. What is the firm's total cost function? Average cost?

***Solution*:**

The variable cost of producing an additional unit, marginal cost, is constant at Birr 500, so

$VC = 500Q$, and $AVC = VC/Q = 500Q/Q = 500$

Fixed cost is 5000 and $AFC = 5000/Q$

$TC = FC + VC$, $TC = 5000 + 500Q$

Average total cost is the sum of average variable cost and average fixed cost:

$ATC = AFC + AVC$, $ATC = 5000/Q + 500$

b. If the firm wanted to minimize the average total cost, would it choose to be very large or very small? Explain.

***Explanation*:** The firm should choose a very large output because average total cost will continue to decrease as Q is increased. As Q becomes infinitely large, ATC will equal Birr 500.

76. Suppose a firm is producing 100 units of output, incurring a total cost of Birr 10 000 and total variable cost of Birr 6000. It can be concluded that average fixed cost is:

***Solution*:** Given: $Q = 100$, $TC = 10{,}000$, $TVC = 6000$, $AFC = ?$

$AFC = TFC/Q$

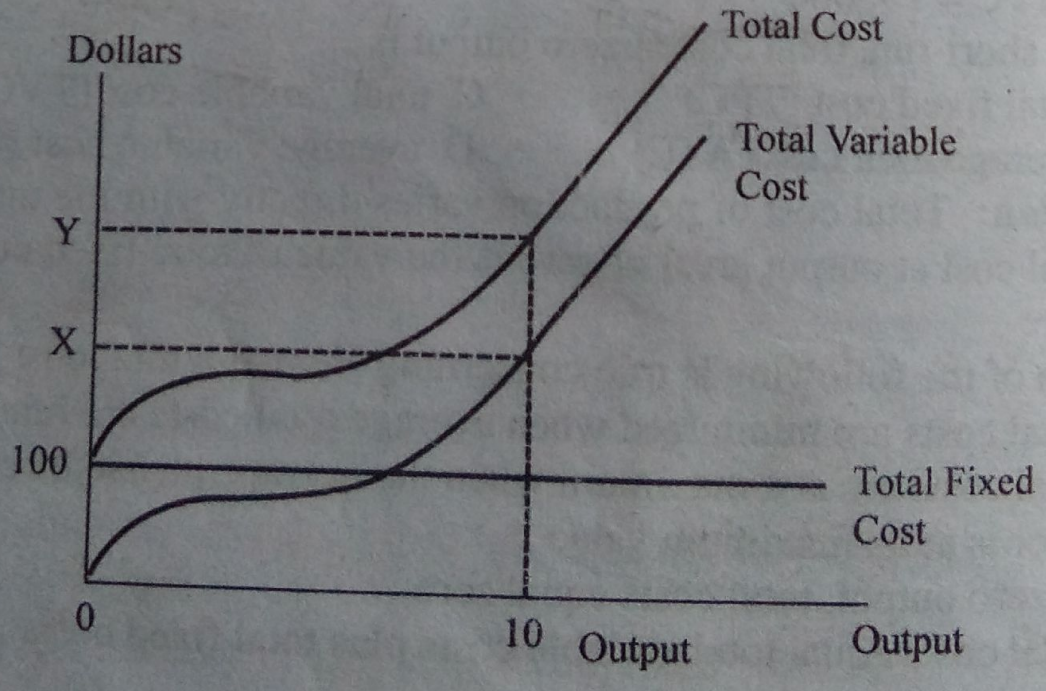
$TC = TFC + TVC$, $TFC = TC - TVC$

$TFC = 10{,}000 - 6{,}000$, $TFC = 4{,}000$

$AFC = TFC/Q$, $AFC = 4{,}000/100$

$AFC = 40$

Questions 77–78 refer to the figure below, which shows cost curves for a competitive firm.



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77. If average variable cost at a quantity of 10 is Birr 25, what is the value of Y in the figure above?
Given: Q = 10, AVC = 25, TFC = 100, TC = ?

Solution:

TC = TFC+TVC
TVC = AVC(Q) = 25 x 10 = 250
TC = TVC+TFC
TC = 250+100= 350

78. At a quantity of 10, what is the value of $(Y - X)$?
Y – is total cost
X- is variable cost
Y – X = fixed cost
FC = TC – TVC
FC= 350 – 250
FC = 100

79. A firm is producing 100 units of output at a total cost of Birr 84,000. The firm's fixed cost is Birr 24,000. What is the average variable cost?

Solution:

Given: Q = 100, TC = 84,000, FC = 24,000
AVC = ?
AC = TC/ Q= 84,000/100
= 840
AFC = TFC/Q = 24,000/100
= 240
AC = AFC+AVC
AVC = AC – AFC
AVC = 840-240
AVC = 600

80. Samson owns a coffee shop that has monthly variable costs equal to Birr 8,560. If Samson sells 11,000 cups of coffee each month, what is the average variable cost per cup?
Given: VC = 8,560, Q = 11.000
AVC = ?

Solution:

AVC = VC/Q
AVC = 8,560/11,000
AVC = 0.78

4.2.3 Average and Marginal Costs in the short-run:

Marginal Cost is the additional cost of producing one more unit of output

MC =ΔTC/ΔQ.

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Since TVC are the only costs that change with the level of output, marginal cost is also calculated as $MC = \Delta TVC/\Delta Q$. If quantity is changing one unit at a time,

$$MC = \Delta TC \Delta Q = \Delta TVC.$$

Average Fixed Cost (AFC) is total fixed cost divided by output.

AFC = TFC/Q. It continuously falls as output rises.

Average Variable Cost (AVC) is total variable cost divided by output.

$$AVC = TVC/Q$$

Average Total Cost (ATC) is total cost divided by output

$$ATC = TC/Q.$$

Note that

$$ATC = AFC + AVC.$$

$$ATC = TC/Q = AVC + AFC$$

Relationship between average costs and marginal cost:

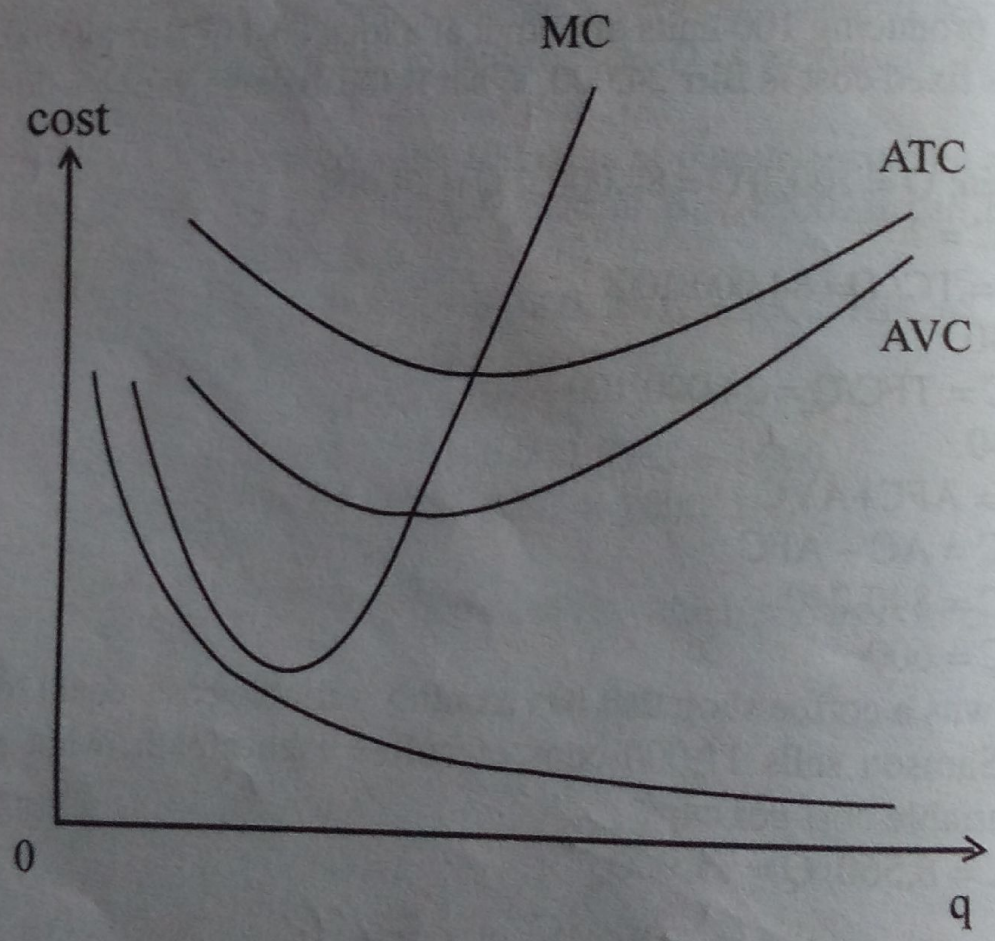

Figure 4.5: average cost and marginal cost.

The relationship between average costs and marginal cost shown in the above diagram

- As output increases, average fixed cost gets smaller and smaller
- As output increases, AVC and ATC curves get closer and closer.
- MC, AVC and ATC curves are all U-shaped
- MC curve intersects AVC and ATC curve at their minimum points

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When MC< AVC, AVC is decreasing;
When MC>AVC, AVC is increasing
When MC< ATC, ATC is decreasing;
When MC> ATC, ATC is increasing

Example: 1. Suppose a firm faces a cost function TC= 10 + 12Q + Q2. Find out:

A. Fixed cost B. Variable cost C. Average cost D. Average variable cost E. Marginal cost

***Solution*:**

A. Fixed cost of the firm does not vary with output level. Fixed cost is 10

B. Total variable cost = total cost minus total fixed cost

$$TVC = TC - TFC$$
$$TVC = (10 + 12Q + Q2) - 10$$
$$TVC = 12Q + Q2$$

C. average cost is total cost per unit

$$AC = TC/Q$$
$$AC = \frac{10 + 12Q + Q_2}{Q}$$
$$AC = 10 + 12 + Q$$

D. Average variable cost is variable cost per unit of output

$$AVC = TVC/Q$$
$$AVC = \frac{12Q + Q_2}{Q}$$
$$AVC = 12 + Q$$

E. Marginal cost = $\Delta TVC / \Delta Q$

$$MC = d\,TVC/dQ$$
$$MC = \frac{d(12Q + Q_2)}{dQ}$$
$$MC = 12 + 2Q$$

Example: 2. Given the following total cost function TC = 200 + 100Q – 12Q 2. Calculate total cost, AC, AVC when the firm produces 8 units of output

***Solution*:**

$$TC = 200 + 100Q - 12Q2$$
$$TC = 200 + 100(8) - 12(8)2$$
$$TC = 200 + 800 - 768$$
$$TC = 232$$

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Average cost = total cost/output

$$AC = TC/Q$$

$$AC = \frac{200 + 100Q - 12Q2}{Q}$$

$$AC = \frac{200}{Q} + 100 - 12Q$$

$$AC = \frac{200}{8} + 100 - 12(8) \qquad AC = 29$$

Average variable cost = Variable cost/ output

$$AVC = TVC/Q$$

$$AVC = \frac{100Q - 12Q2}{Q}$$

$$AVC = 100 - 12(8) \qquad AVC = 4$$

Illustrative Examples

81. Which of the following could shift a firm's average variable cost curve upward?
    A. A decrease in the productivity of labor.
    B. An increase in the productivity of labor. C. An increase in fixed cost.
    D. A decrease in the demand for the good produced by that firm.

Explanation:

- Average Variable Cost (**AVC**) changes only when Marginal Cost (MC) changes, and vice-versa. **AVC** and **MC** are both pure variable cost measures
- Labor is usually considered a variable input. As labor becomes less productive **MC** and **AVC** increase.
- A change in demand for a product doesn't affect the production costs curves. It may affect how much is actually produced and so affect where on the cost curve the firm operations, but it doesn't shift the curves.
- AVC and fixed cost are unrelated
- More productive labor causes MC and AVC to decrease or shift downward.

Answer: A

82. Which of the following relationships is NOT valid?
    A. Rising marginal cost implies that average total cost is also rising.
    B. When marginal cost is below average total cost, the latter is falling.
    C. When marginal cost is above average variable cost, AVC is rising.
    D. none of the above

Explanation:

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- When MC < AC, AC declines
- When MC > AC, AC rises
- When MC = AC, MC reaches its minimum

Relationship between MC and AVC are similar to the above relationship.

Answer: A

83. Average fixed cost:

A. is U-shaped
B. decline over the entire output range
C. is a long run concept
D. is influenced by diminishing returns

***Explanation*:** Average fixed cost declines continuously as the firm increases its output level

Answer: B

84. If average fixed cost is 40 and average variable cost is 80 for a given output, what is average total cost?

Given: AFC= 40, AVC= 80

***Solution*:** Average cost = average fixed cost + average variable cost

$$AC = AFC + AVC$$
$$AC = 40 + 80$$
$$AC = 120$$

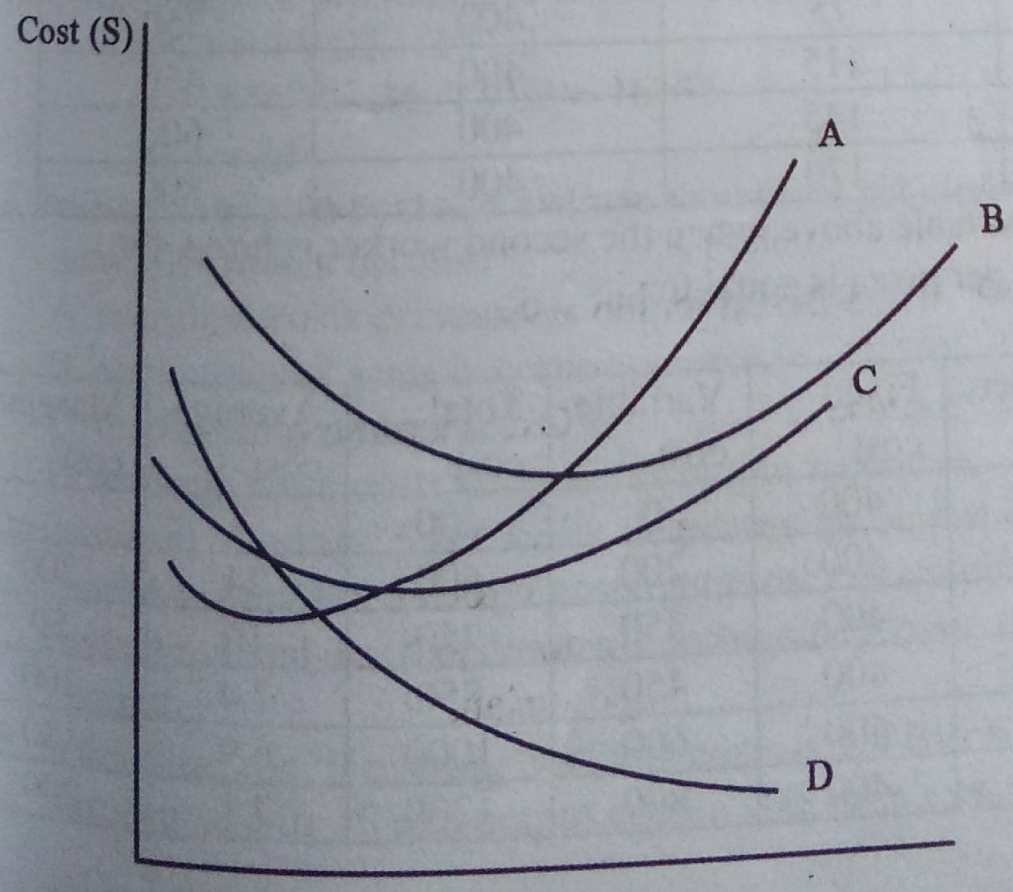
85. The slope of the total variable cost curve equals:

A. Average variable cost
C. average cost
B. Marginal cost
D. marginal physical product

***Explanation*:** Average variable cost equals total variable cost divided by output level $AVC = TVC/Q$

The slope of the total variable cost curve is average variable cost.

Answer: A



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The set of lines above reflect information about the cost structure of a firm. Use the figure to answer the following questions.

86. Refer to the above Graph. Which of the lines is most likely to represent average variable cost?

A. A B. B C. C D. D

Explanation: Line B represents average variable cost **Answer: C**

87. Refer to the above Graph. Which of the lines is most likely to represent marginal cost?

A. A B. B C. C D. D

Explanation: Line C represents marginal cost **Answer: A**

88. Suppose that a firm incurs a total cost of Birr 974 when it produces 10 units and a total cost of Birr 1060 when it produces 11 units. It can be concluded that the marginal cost of producing the eleventh unit is

Given: TC1 = 974, TC2 = 1068

Q1= 10, Q2= 11

Solution:

MCn = TCn – TCn-1

MC11= TC11 – TC 10

MC= 1068 – 974

MC = 94

Question 89, 90 and 91 are based on the following table below.

Workers per day	Pizza per day	Fixed cost in Birr per day	Variable cost in Birr per
0	0	400	0
1	25	400	200
2	75	400	350
3	115	400	450
4	145	400	600
5	170	400	800

89. Referring to the table above, when the second worker is hired, the marginal cost per pizza is equal to:

Solution:

Workers per day	Pizza per day	Fixed cost	Variable cost	Total cost	Average cost	Marginal cost
0	0	400	0	400	-	-
1	25	400	200	600	24	200
2	75	400	350	750	10	50
3	115	400	450	850	7.4	100
4	145	400	600	1000	6.9	150
5	170	400	800	1200	7.1	200

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• When the second worker is hired, the marginal cost per pizza is Birr 50

90. Referring to the table above, the total average cost of producing 25 pizzas a day is:

Explanation:

• Based on the above table, the total average cost of producing 25 pizzas per a day is 24.

91. Again referring to the table above, assuming 5 workers hired per day, the total cost of producing pizzas per day is:

Explanation:

• The total cost of producing pizzas hiring 5 workers per day is 1200

92. In the short run, the marginal cost of the first unit of output is Birr 20, the marginal cost of producing the second unit of output is Birr 16, and the marginal cost of producing the third unit of output is Birr 12. The firm's total variable cost of producing three units of output is:

Solution:

Given: MC1 = 20, MC2 = 16, MC3 = 12

Total variable cost producing of the first unit of output is:

MC = TVCn – TVCn-1

MC1= TVC1 – TVC0

20 = TVC1- 0

TVC1 = 20

Total variable cost producing of the 2nd unit of output is:

MC2= TVC2 – TVC1

16 = TVC2 – 20, TVC2 = 16+20

TVC2 = 36

Total variable cost producing of the 3rd unit of output is:

MC3 = TVC3 – TVC2

16 = TVC3 – 36, TVC3 = 16+36

TV3 = 48

93. Average variable cost and average total costs get closer together as output increases because

A. marginal costs decrease as output increases.

B. economies of scale become apparent.

C. diminishing returns set in.

D. average fixed costs decrease as output increases.

Explanation: Average fixed costs decreases as output increases. Average variable cost and average costs get closer as output increases because average fixed cost decreases continuously over the entire range of output.

Answer: D

94. In the short run, if average variable costs equal Birr 6 and average total costs equal Birr 10 and output equals 100, then total fixed costs equal:

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Given: AVC= 6, AC = 10, Q = 100

***Solution*:**

TFC= TC - TVC

AVC = TVC/ Q

TVC= AVC x Q = 6x100

TVC = 600

AC = TC/Q

TC= AC x Q = 100 x 10

TC = 1000

TFC = TC – TVC

TFC = 1000 – 600

TFC = 400

95. At its minimum point, the average total cost curve is intersected by the
    A. total fixed cost curve
    B. total variable cost curve
    C. average fixed cost curve
    D. marginal cost curve

***Explanation*:** Marginal cost curve cuts average variable cost and average cost curve at their minimum level.

Answer: D

96. Which of the following could be used to compute ATC?
    A. (TVC - TFC)/Output
    B. (AFC + AVC)/Output
    C. TC/Input
    D. (TFC + TVC)/Output

***Explanation*:**

Average cost = total cost/ output

TC = TFC + TVC

AC= TC/Q = TFC+ TVC/Q

AC = TFC/Q + TVC/Q

AC = AFC + AVC

Answer: D

97. If marginal cost is increasing but less than average total cost, what do we know about average total cost?
    A. Average total cost is decreasing.
    B. Average total cost is constant.
    C. Average total cost is increasing.
    D. Any of the above could be true.

***Explanation*:**

- When marginal cost equals average cost, then average cost reaches its minimum level.
- When marginal cost increases but less than average cost, then average cost decreases.
- When marginal cost increases and greater than average cost, then average cost increases.

Answer: A

98. Which of the following cost and production relationships is inaccurately stated?

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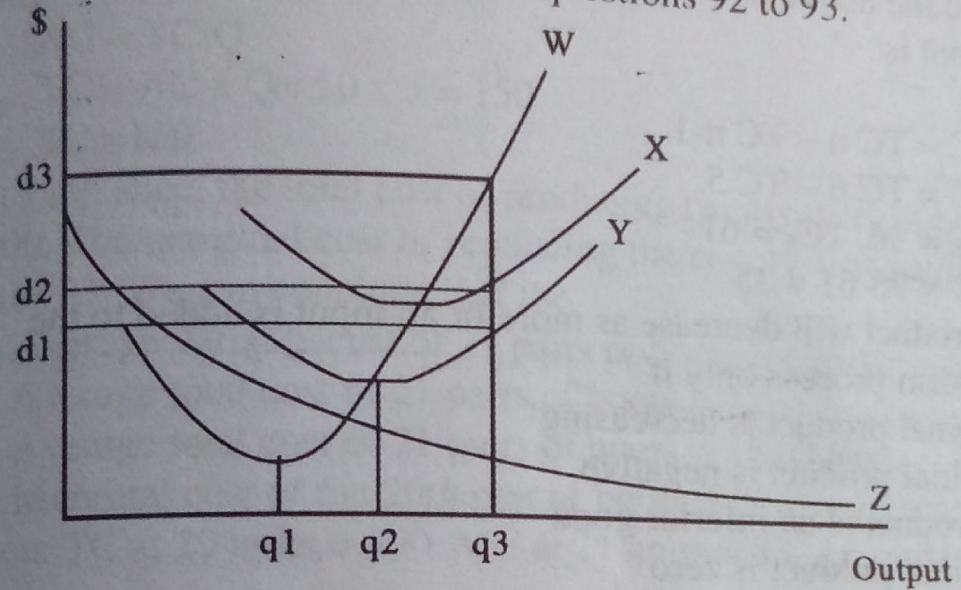
A. AFC = AVC – ATC
B. MC = ΔTVC/ΔQ
C. TVC = TC – TFC
D. APL = TPL/L
E. MC = w/MPL

Explanation:

- AFC plus AVC equals ATC. If you do the subtraction, AFC= ATC–AVC,
- If you have studied your production and cost relationships, you recognize that choices B, C, D and E are all stated correctly.

Answer: A

Use the following figure to respond to questions 92 to 93.



99. The curves labeled W, X, Y, Z refer to which respective cost functions?
A. MC, AVC, ATC, and AFC.
B. MC, TC, TVC, and AFC.
C. MC, ATC, AVC, and AFC.
D. MC, ATC, AVC, and TFC.

Explanation: The curve W- marginal cost (MC), X is average total cost(AC), Y is average variable cost(AVC) and Z is average fixed cost(AFC)

Answer: C

100. At the *q*3 level of output,
A. AFC = d2 – d1.
B. MC = d2.
C. TVC = d2.
D. ATC = d3.
E. AFC = d3 – d2.

Explanation: The vertical distance between ATC and AVC is AFC at any level of output.

Answer: A

Output	0	1	2	3	4	5	6
Total cost	24	33	41	48	54	61	96

101. Refer to the above data. The total variable cost of producing 5 units is:

Solution:

TC = TVC + TFC, TVC = TC - TFC
TFC = 24
At 5 units of out TC = 61
TVC = TC –TFC

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TVC = 61 – 24 = 37

102. Refer to the above data. The average total cost of producing 3 units of output is:

Solution:

ATC at 3 units of output = 48

ATC = TC/Q

ATC = 48/3 = 16

103. Refer to the above data. The marginal cost of producing the sixth unit of output is:

Solution:

MC = TC n – TC n-1

MC = TC 6 – TC 5

$TC_6 = 96$, $TC_5 = 61$

MC = 96-61 = 35

104. Total product will decrease as more of an input is added to the production process only if

A. marginal product is decreasing
B. marginal product is negative
C. the product is an inferior good
D. marginal product is zero

Explanation:

- Marginal product is the change in total product that occurs when the usage of a particular resource increases by one unit, all other resources constant.
- For example, if a firm hires one more worker and its total product increases, marginal product is positive; if its total product declines when an additional worker is hired, the marginal product is negative.

Answer: B

105. If short-run marginal cost is initially declining due to the benefits of specialization of labour, the average variable cost will

A. Decline and be below marginal cost at the initial output levels.
B. Decline and be the same as marginal cost at the initial output levels.
C. Decline and be above marginal cost at the initial output levels.
D. Rise and be below marginal cost at the initial output levels.

Explanation:

- When marginal cost is less than either average variable cost or average cost, it causes them to decrease.
- When marginal cost is above average cost or average variable cost, it causes them to increase.
- When marginal cost decline, average variable cost decline and be above marginal cost at the initial output levels

Answer: C

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106. Suppose that for a particular firm the only variable input into the production process is labor and that output equals zero when no workers are hired. In addition, suppose that marginal cost of the third worker hired is 40, and the average total cost when three workers are hired is 50. What is the total cost of production when three workers are hired?

Given: MC = 40, AC = 50, Q = 3, TC = ?

Solution:

AC = TC/Q

TC = AC x Q= 50 x 3 = 150

TC =150

107. At Lion shoe, the total cost of producing twenty pairs of boots is Birr 400. The marginal cost of producing the twenty-first pair of boots is Birr 83. We can conclude that the

A. Average variable cost of 21 pairs of boots is 23 Birr.
B. Average total cost of 21 pairs of boots is 23
C. Average total cost of 21 pairs of boots is 15.09 Birr.
D. Marginal cost of the 20th pair of boots is 20 Birr.

Given: TC at 20 units = 400, MC at 21 units = 83, AC at 21 units = ?

Solution:

MC = TC n – TC n-1

MC21 = TC 21 – TC 20

83 = TC 21 – 400

TC 21 = 83 +400

TC 21 = 483

AC at 21 units = TC/Q

AC = 483/ 21

AC = 23

Answer: B

108. Suppose that for a particular firm the only variable input into the production process is labor and that output equals zero when no workers are hired. In addition, suppose that the average total cost when 5 units of output are produced is 30, and the marginal cost of the sixth unit of output is

60. What is the average total cost when six units are produced?

Solution:

Given: AC = 30 at 5 units, MC = 60 at 6 units, AC = at 6 units?

Solution:

AC = TC/Q

TC at 5 units = 30 x 5 = 150

MC = TC n – TC n-1

MC 6 = TC 6 – TC 6-1= TC 6 – TC5

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60 = TC6 – 150
TC 6 = 60 + 150 = 210
AC at 6 units = TC/Q = 210/6
AC at 6 units = 35

Question 109 – 111are based on the information about a cost given below

Q	AFC	AVC
1	60	100
2	30	90
3	20	60
4	15	50
5	12	60
6	10	70
7	8.6	80
8	7.5	90

109. What is the firm fixed cost?

***Solution*:**

FC = AFC x Q
FC = 60 x 1 = 60
FC = 2x 30 = 60
FC = 8 x 7.5 = 60
The firms' fixed cost is 60

110. If the firm produces four units, what is the average total cost?

***Solution*:**

At four units, AFC = 15, AVC = 50
AC = AFC + AVC
AC = 15+ 50 = 65

111. What is the total cost of producing five units?

***Solution*:**

At five units, AC = AFC + AC = 12 + 60 = 72
AC = TC/Q
TC = AC x TC = 72 x 5
TC = 360
Or
TFC = 60
TVC = AVC x Q = 60 x 5 = 300
TC = TFC + TVC
TC = 60 + 300 = 360

112. The shape of the long-run average cost curve reflects
A. market demand
B. economies and diseconomies of scale
C. increasing and diminishing marginal returns D. all of the above

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Explanation:

- An economy of scale occurs when forces cause a reduction in a firm's average cost as the scale of operations increases in the long-run.
- A diseconomy of scale occurs when forces cause a firm's average cost to increase as the scale of operations increases in the long-run.

Answer: B

113. Suppose the total cost of producing T-shirts can be represented as TC=50+2Q. The marginal cost of the 5th T-shirt is

Solution:

TC = 50 +2Q
MC = TC n – TC n-1
MC= TC5 – TC4
MC = 50+2(5) – 50+ 2(4) = 60 – 58 = 2
MC of the 5th unit is 2

114. Suppose the total cost of producing T-shirts can be represented as TC=50+2Q. The average cost of producing 5 T-shirts

Solution:

AC = TC/ Q
TC = 50+ 2(5) = 60
AC = 60/5 = 12
OR
AC = TC/Q

$$AC=\frac{50+2Q}{Q}=\frac{50}{Q}+\frac{20}{Q}+\frac{50}{5}+2=12$$

115. Why does the difference between Average Total Cost and Average Variable Cost decrease with an increase in the level of output? Can these two be equal at some level of output?

Explanation

- Average Total Cost (ATC) minus Average Variable Cost (AVC) is equal to Average Fixed Cost (AFC).
- AFC = TFC / Output.
- As the level of output increases, AFC falls. So, the difference between ATC and AVC decreases with increase in output.

ATC and AVC can never be equal at any level of output as AFC can never be zero because TFC is positive.

116. When an increasing quantity of a variable factor is used with a set of fixed inputs, the marginal productivity of the variable input will:

A. Continuously increase
B. First decline and eventually increase
C. First increase and eventually decline

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D. First increase and eventually remain constant

***Explanation*:**

- The marginal product of labour is the additional output produced by a firm as a result of hiring one more worker.
- Specialization and division of labour cause the marginal product of labour to rise for the first few workers hired.
- The law of diminishing returns causes the marginal product of labour to decline.

Answer: C

117. Which of the following cost and production relationships is inaccurately stated?

A. AFC = AVC –ATC B. MC = ΔTVC/ΔQ C. TVC = TC –TFC
D. APL = TPL/L E. MC = w/MPL

***Explanation*:**

ATC = AFC + AVC AFC = ATC – AVC

Answer: A

4.2.4 Relationship between production and cost:

Marginal product and Marginal cost:

The U-shape of the marginal cost curve is closely related to the hump-shape of the marginal product curve.

- The increasing portion of the marginal product curve corresponds with the decreasing portion of the marginal cost curve.
- The decreasing portion of the marginal product curve corresponds with the increasing portion of the marginal cost curve.
- The peak of the marginal product curve corresponds with the minimum of the marginal cost curve.

Because variable cost is largely associated with the cost of employing a variable input in the short run, it is possible to identify a connection between the marginal cost curve and the marginal product curve.

- The quantity of output in which marginal cost is at a minimum is the same quantity of output produced by the variable input when the marginal product of the variable input is at a maximum.
- Over the range of production in which marginal product increases and the variable input experiences increasing marginal returns, the marginal cost curve declines.
- Over the range of production in which marginal product increases and the variable input experiences decreasing marginal returns brought on by the law of diminishing marginal returns, the marginal cost curve is rising.

The correspondence between the marginal product and marginal cost curves indicates that the law of diminishing marginal returns is the key reason for increasing marginal cost.

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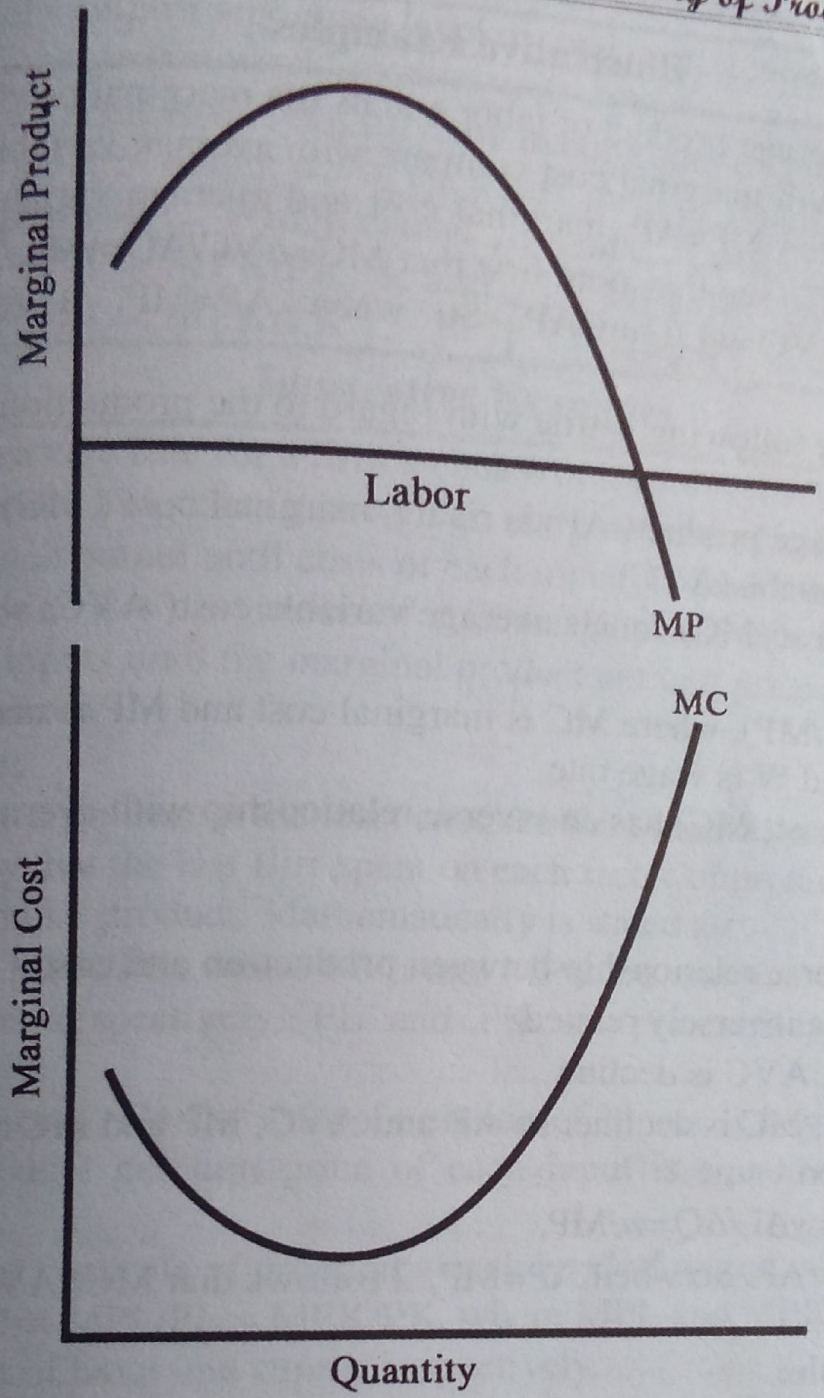

Figure 4.5: relationship between marginal product and marginal cost.
The marginal product curve in the upper panel has a distinctive hump-shape, with marginal product rising, reaching a peak, and then falling.

- The rising portion of the marginal product curve is the result of increasing marginal returns.
- The falling portion is attributable to decreasing marginal returns, and in particular, the law of diminishing marginal returns.

The marginal cost curve in the lower panel has a distinctive U-shape, with marginal cost falling, reaching a minimum, and then rising.

- The falling portion of the marginal product curve is the result of increasing marginal returns.
- The rising portion is attributable to decreasing marginal returns, and in particular, the law of diminishing marginal returns.

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118. When the average product of labor equals the marginal product of labor, how will marginal cost compare with average variable cost?

***Explanation*:** When MP=AP, marginal cost and average variable cost will be the same. To see this, note first that MC=ΔVC/ΔQ=wΔL/ΔQ=w/MP. Similarly, AVC=wL/Q=w/AP. So when AP=MP, it follows that MC=AVC.

119. Which of the following is true with regard to the production- cost linkage?

A. When average product(AP) is rising, marginal cost (MC) is above average variable(AVC)

B. Marginal cost(MC) equals average variable cost(AVC) when AVC is rising

C. MC = W(1/MP), where MC is marginal cost and MP is marginal product and W is wage rate

D. Marginal cost(MC) has an inverse relationship with average product (AP)

***Explanation*:**

- There is an inverse relationship between production and cost
- TVC and TP are inversely related
- When AP rises, AVC is decline
- When MP rises, MC is decline, so AP and AVC, MP and MC are inversely related
- MC=ΔVC/ΔQ=wΔL/ΔQ=w/MP.
- AVC=wL/Q=w/AP. So when AP=MP, it follows that MC=AVC.

Answer:

The least cost rule:

That is the cost of any output is minimized when the ratios of marginal product to price of the last units of resources used are the same for each resource.

$$MPL/PL = MPK/PK = PL/PK$$

In situations like this,

- Where MPL/PL > MPK/PK, the firm is going to find it in its best interest to increase spending on L and decrease spending on K.
- The Law of Diminishing Marginal Returns predicts that as you increase L, MPL falls. And as you decrease K, MPK rises. The substitution of labor for capital ceases to be a great deal at the combination of L and K where the ratios of marginal product per dollar are equal again.
- For any given level of total output, a profit maximizing firm would want to minimize its cost.

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- Alternatively put, for any given level of total cost, it would want to choose the mix of inputs that would maximize its output.
- The firm can achieve this objective by making sure that it is getting the highest amount of output out of each Birr spent on inputs.

Situation	Firm will ...	Which causes ...	And ...	Until
MPL/PL > MPK/PK	↑L	and	↓ ↓	↑ MPL/PL = MPK/PK
MPL/PL < MPK/PK	↑	↓	↓	↑ MPL/PL = MPK/PK

Illustrative Examples

120. The least cost rule for a firm's choice of inputs is that it should:
   A. Combined inputs that have high marginal rate of substitution
   B. Produce output until costs of each input purchased is equalized
   C. Combine low cost inputs of high marginal rate of substitution
   D. Buy inputs until the marginal product per unit price of each input is equalized

Explanation:

- To produce a product at the least cost, the firm should spend its money in such a way that the last Birr spent on each factor of production brings equal marginal product. Mathematically is stated as:
- MPL/PL= MPK/PK, where MPL and MPK are marginal product of labor and capital respectively, PL and PK are price of labor and capital respectively

So the least cost rule for a firm's choice of inputs is buy inputs until the marginal product per unit price of each input is equalized (***MPL/PL = MPK/PK)*** **Answer: D**

121. The least cost rule of producing each level of output in the long run states that MPL/PL = MPK/PK, where MPL and MPK are marginal product of labor and capital respectively.

PL and PK are price of labor and capital respectively. If a particular producer observes that MPL/PL > MPK/PK he/ she should _____ , in order to minimize cost.

   A. Reduce both labor and capital inputs
   B. Increase both labor and capital inputs
   C. Reduce labor inputs and increase capital inputs
   D. Increase labor inputs and reduce capital inputs

Explanation:

- If a particular producer observes that MPL/PL > MPK /PK, the producer is not in equilibrium because the marginal product of labor per unit price is not equal to the marginal product of capital per unit price
- The producer should continue to substitute labor for capital until the equality holds. This is because the quantity of labor increases and

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quantity of capital decreases to produce the same level of output the MPL decreases and MPK increases until equality is established

- In order to minimize cost of the producer should reduce capital input and increase labor input

Answer: D

122. Given: Q = 5 KL

Where Q is the number of output, K is the number of capitals, and L is the number of labor. The price of capital r = 10,000 per week and the price of labor w = 5,000 per week. Output costs are given by the cost of labor and capital, plus Birr 2,000 per output for raw materials. Your plant has a fixed installation of 5 capital machines as part of its design.

A. What is the cost function for your plant — namely, how much would it cost to produce *output*? What are average and marginal costs for producing Q? How do average costs vary with output?

K is fixed at 5. The short-run production function then becomes Q = 25L. This implies that for any level of output Q, the number of labor teams hired will be L = Q/25. The total cost function is thus given by the sum of the costs of capital, labor, and raw materials

$$TC(Q) = rK + wL + 2000Q = 1000x5 + 5000(Q/25) + 2000Q$$

$$TC(Q) = 50{,}000 + 2200Q$$

The average cost function is given by:

$$AC = TC/Q$$

$$AC = \frac{50{,}000 + 2200Q}{Q}$$

and the marginal cost function is given by:

$$MC(Q) = dTC/dQ$$

$$MC = 2200$$

Marginal costs are constant and average costs will decrease as quantity increases (due to the fixed cost of capital).

B. How many teams are required to produce 250 outputs? What is the average cost per output?

To produce Q = 250 output we need labor teams L= Q/25, L= 250/25, L= 10

Average costs are given by:

$$AC = \frac{50{,}000 + 2200Q}{Q}$$

$$AC = \frac{50{,}000 - 2200(250)}{250}$$

$$AC = 2400$$

C. You are asked to make recommendations for the design of a new production facility.

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What capital/labor *(K/L)* ratio should the new plant accommodate if it wants to minimize the total cost of producing any level of output Q?

We no longer assume that K is fixed at 5. We need to find the combination of K and L that minimizes costs at any level of output q. The cost-minimization rule is given by:

$$\frac{MP_K}{r} = \frac{MP_L}{w}$$

To find the marginal product of capital, observe that increasing K by 1 unit increases q by 5L, so MPK = 5L. Similarly, observe that increasing L by 1 unit increases Q by 5K, so MPL = 5K. Mathematically,

MPK = dQ/ dK= 5L

MPL = dQ/ dL =5K.

Using these formulas in the cost-minimization rule, we obtain:

5L/r = 5K/L

K/L = w/r

5000/10,000 = 1/2.

The new plant should accommodate a capital to labor ratio of 1 to 2.

123. A firm's production function is Q = min (K, 2L), where Q is the number of units of output produced using K units of capital and L units of labor. The factor prices are w = 4 (for labor) and r = 1 (for capital) calculate the total cost at Q = 12

***Solution*:** Given: Q = 12, w = 4, r = 1, TC =?

Q= (K , 2L)

12 = (K , 2L)

To produce Q = 12, the firm will need at least K = 12 and L = 6.

TC = wL + rK

TC = 4x6 + 1x12, TC= 36

124. A firm employs labor as the only variable factor along with a fixed quantity of capital. Wage rate of labor is 60 per day and its marginal cost is 30 units of output. What is marginal product of labor?

***Solution*:** Given: w = 60, MC= 30, MPL = ?

MC = w/ MPL

MPL = w/ MC, MPL = 60/ 30

MPL = 2

125. Given [MPL, PL] and [MPK , PK] respectively, where, MPL is the marginal product of labour, PL s price of labour, MPK is the marginal product of capital, PK is price of capital.

Which of the following is a least production state of a firm?

A. [100, 25] and [100, 50]

B. [100, 50] and [200, 100]

C. [200, 50] and [100, 75]

D. [200, 100] and [100, 2000]

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Explanation: Least cost rule is:

MPL/PL = MPK/PK

100/50 = 200/100

2=2

Supplementary Exercise on Unit 4

1. Suppose total revenues = 100,000 while explicit costs=40,000 and implicit costs/opportunity costs=10,000. Compute accounting and economic profit.
   A. The economic profit = 10,000
   B. The economic profit = 40,000
   C. the economic profit = 50,000
   D. The economic profit = 60,000
2. In the short run, a firm's fixed cost
   A is zero.
   B. cannot be escaped.
   C. can be escaped only by cutting production to zero.
   D. is not correctly described by any of the above.
3. The vertical distance between total cost curve and total variable cost curve is equal to
   A. average fixed cost.
   B. total fixed cost.
   C. average variable cost.
   D. average total cost
4. The long run is a time period of sufficient length to enable
   A. producers to alter their use of fixed capital (the size of their plant and equipment).
   B. producers to alter their output by utilizing labor and raw materials more intensively.
   C. decision makers to adjust fully to a change in market conditions.
   D. Both A and C are correct
5. The short run is a period of time
   A. during which all inputs are fixed
   B. that is just long enough to permit entry and exit
   C. that is always less than one year
   D. during which at least one input is fixed
6. For a short-run cost function which of the following statements is (are) **not** true?
   A. The average fixed cost function is monotonically decreasing.
   B. The marginal cost function intersects the average fixed cost function where the average variable cost function is a minimum.
   C. The marginal cost function intersects the average variable cost function where the average variable cost function is a minimum.
   D. The marginal cost function intersects the average total cost function where the average total cost function is a minimum
7. Which of the following is part of a firm's explicit costs?

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A. wages
B. interest on a bank loan
C. interest forgone on funds used to buy capital equipment D. All

8. An implicit cost is an opportunity cost that
   A. is actually part of the firm's normal profit.
   B. is measured by the amount of cash the firm actually pays out.
   C. is adjusted for the rate of inflation.
   D. requires no actual payment of cash.
9. Most typically, a firm incurs an implicit cost when it
   A. uses its capital equipment.
   B. uses labor.
   C. pays for its utilities, such as electric power.
   D. pays interest on a loan to a bank.
10. Which of the following is most likely to be an implicit cost of production?
   A. property taxes on a building owned by the firm
   B. transportation costs paid to a trucking supplier
   C. rental payments for a building utilized by the company and rented from another party
   D. interest income foregone on funds invested in the firm by the owners
11. Average total cost is equal to:
   A. average fixed cost + average variable cost.
   B. total cost - total quantity of output.
   C. average variable cost + total fixed cost.
   D. average variable cost x total quantity of output.
12. The output where diminishing returns to production begin is also the output where
   A. marginal cost is at a minimum.
   B. average total cost is at a minimum.
   C. average variable cots is at a minimum.
   D. marginal and average cost intersect.
13. Which of the following statements about marginal cost is ***incorrect***?
   A. A U-shaped marginal cost curve implies the existence of diminishing returns over all ranges of output.
   B. When marginal cost equals average cost, average cost is at its minimum.
   C. In the short run, the shape of the marginal cost curve is due to the law of diminishing marginal returns.
   D. When marginal cost is falling, total cost is rising.

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14. Which of the following statements about the relationship between marginal cost and average cost is ***correct***?
    A. When MC is falling, AC is falling.
    B. AC equals MC and MC's lowest point.
    C. When MC exceeds AC, AC must be rising.
    D. When AC exceeds MC, MC must be rising.
15. If marginal cost is below average total cost, average total cost
    A. is falling. B. is rising. C. is constant.
    D. may rise or fall depending on the size of fixed costs.
16. Which one of the following statements is FALSE?
    A. TC = TFC + TVC C. AFC = TFC divided by Q
    B. ATC = AFC + AVC D. MC = TC divided by Q
17. In the short run, average total cost is
    A. higher than average variable cost.
    B. equal to average variable cost.
    C. less than average variable cost.
    D. sometimes higher and sometimes lower than average variable cost.
18. The amount that total cost raises when the firm produces one additional unit is called
    A. marginal cost. C. fixed cost
    B. average cost. D. variable cost
19. Average total cost tells us
    A. the cost of a typical unit of output, if total cost is divided evenly over all the units produced.
    B. the cost of the last unit of output, if total cost does not include a fixed cost component.
    C. the variable cost of a firm that is producing at least one unit of output.
    D. the total cost of the first unit of output, if total cost is divided evenly over all the of all resources used in a production process.
20. Increasing marginal costs necessarily mean that
    A. average total cost is increasing
    B. average variable cost is decreasing
    C. average fixed cost is constant
    D. average total cost may be increasing or decreasing
21. At the output level where MC is equal to ATC,
    A. MR is equal to MC
    B. MC is also equal to AVC
    C. they are both less than AFC
    D. ATC is at its minimum value
22. An isoquant curve shows

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A. all the alternative combinations of two inputs that yield the same maximum total product.
B. all the alternative combinations of two products that can be produced by using a given set of inputs fully and in the best possible way.
C. all the alternative combinations of two products among which a producer is indifferent because they yield the same profit.
D. both B and C.

23. A negatively sloped isoquant implies
A. products with negative marginal utilities.
B. products with positive marginal utilities.
C. inputs with negative marginal products.
D. inputs with positive marginal products.

24. Which of the following are characteristics of a typical isoquant?
A. All input combinations on the isoquant will produce the same level of output
B. A change in inputs prices shifts the isoquant map
C. The marginal rate of technical substitution decreases as labor is substituted for capital by moving down the isoquant
D. Both A and C

25. A firm experiences diminishing marginal returns only when:
A. Its total product is decreasing.
B. Its average product is decreasing.
C. Its marginal product is decreasing.
D. Its marginal product is decreasing but greater than zero.

26. Economies of scale occur when the average total cost of production as output.
A. increases, increases
B. decreases, decreases
C. decreases, increases
D. remains constant, increases

27. An isocost line reveals the:
A. costs of inputs needed to produce along an isoquant.
B. costs of inputs needed to produce along an expansion path.
C. input combinations that can be purchased with a given outlay of funds.
D. output combinations that can be produced with a given outlay of funds.

28. Lines that represent combinations of inputs that cost the same total amount are called
A. Isoquants B. Isocost lines C. Profit curves D. Cost curves

29. An isocost line will be shifted further away from the origin
A. If the prices of both inputs increases
B. If total cost increases
C. If there is an advance in technology
D. All of the above are correct

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30. A firm achieves producer equilibrium--- where costs are minimized/output is maximized when:
    A. The isoquant is tangent to the isocost line C. The MRTSLK = -PL/PK
    B. MPL/PL = MPK /Pk D. All the above are correct
31. An isocost line identifies
    A. the least costly combination of inputs needed to produce a given level of output.
    B. the relative prices of inputs.
    C. the technological relationships among inputs.
    D. the rate at which one input can be substituted for another in the production process.
32. For a given short-run production function,
    A. technology is assumed to change as capital stock changes.
    B. technology is assumed to change as the labor input changes.
    C. technology is considered to be constant for a given production function relationship.
    D. technology is assumed to change positively until diminishing returns set in and then it changes in the other direction.
33. A firm in the short run:
    A. Cannot have decreasing marginal product as input is increased.
    B. Cannot have decreasing average product as input is increased
    C. Cannot have increasing average product as input is increased.
    D. Cannot increase its plant size.
34. In the short run, production is explained by
    A. The law of variable proportions C. The law of returns to scale
    B. Economies and diseconomies of scale
    D. The law of diminishing marginal utility
35. The marginal rate of technical substitution is
    A. the rate at which a producer is able to exchange, without affecting the quantity of output produced, a little bit of one input for a little bit of another input.
    B. the rate at which a producer is able to exchange, without affecting the total cost of inputs, a little bit of one input for a little bit of another input.
    C. the rate at which a producer is able to exchange, without affecting the total inputs used, a little bit of one output for a little bit of another output.
    D. a measure of the ease or difficulty with which a producer can substitute one technique of production for another.
36. In any production process the marginal product of labor equals:
    A. total output divided by total labor inputs.

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B. total output minus the total capital stock .
C. total output produced by labor inputs
D. the change in total output resulting from a 'small' change on the labor input

37. If average variable cost of producing 120 units of output is 60 and total fixed cost is 60,000. Determine total cost
38. Given a total cost functions C = 120 + 200X2 – 2X2 +4X3, find
A. TVC when X = 2 units
B. AVC when X =1 units
C. AC when X = 4 units
D. FC when X = 6 units
E. AFC when X= 2 units
F. MC when X= 2 units

Answer to Supplementary Exercise on Unit 4

1. C	2. B	3. B	4. D	5. D	6. B	7. D	8. D	9. A	10. D
11. A	2. A	13. D	14. C	15. A	16. D	17. A	18. A	19. A	20. A
21. D	22. A	23. C	24. D	25. D	26. C	27. C	28. B	29. B	30. D
31. A	32. C	33. D	34. A	35. D	36. D	37. 67200			

38. A) 824 B) 202 C) 886 D) 120 p E) 60 F) 840

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